

Linear Algebra for Computer Science

(Línuleg algebra með
tölvunarfræði)

Lecture 11: The Basis

Instructors:

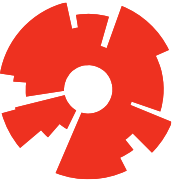
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[Based on slides for „Decoding the Matrix“]



WEEK

SIX



Agenda

Chapter 5

Lecture 11:

- Coordinate Systems
- Coordinate Representation
- Finding a Set of Generators
- Grow Algorithm
- Shrink Algorithm
- Minimum Spanning Forest
- Superfluity
- Linear Dependence
- Linear (In)dependence Properties





Recap: Some Key Take-aways (Week 5)

A **linear function** can be expressed as a matrix-vector multiplication and has two properties (L1, L2)

Link of the Matrix to Linear Functions

The **kernel of a linear function** is the set of inputs that are mapped by the function to the zero vector

Similar to Null Space of a Matrix

A vector can be interpreted as a **column vector** meaning a one-column matrix or a **row vector** meaning a one-row matrix

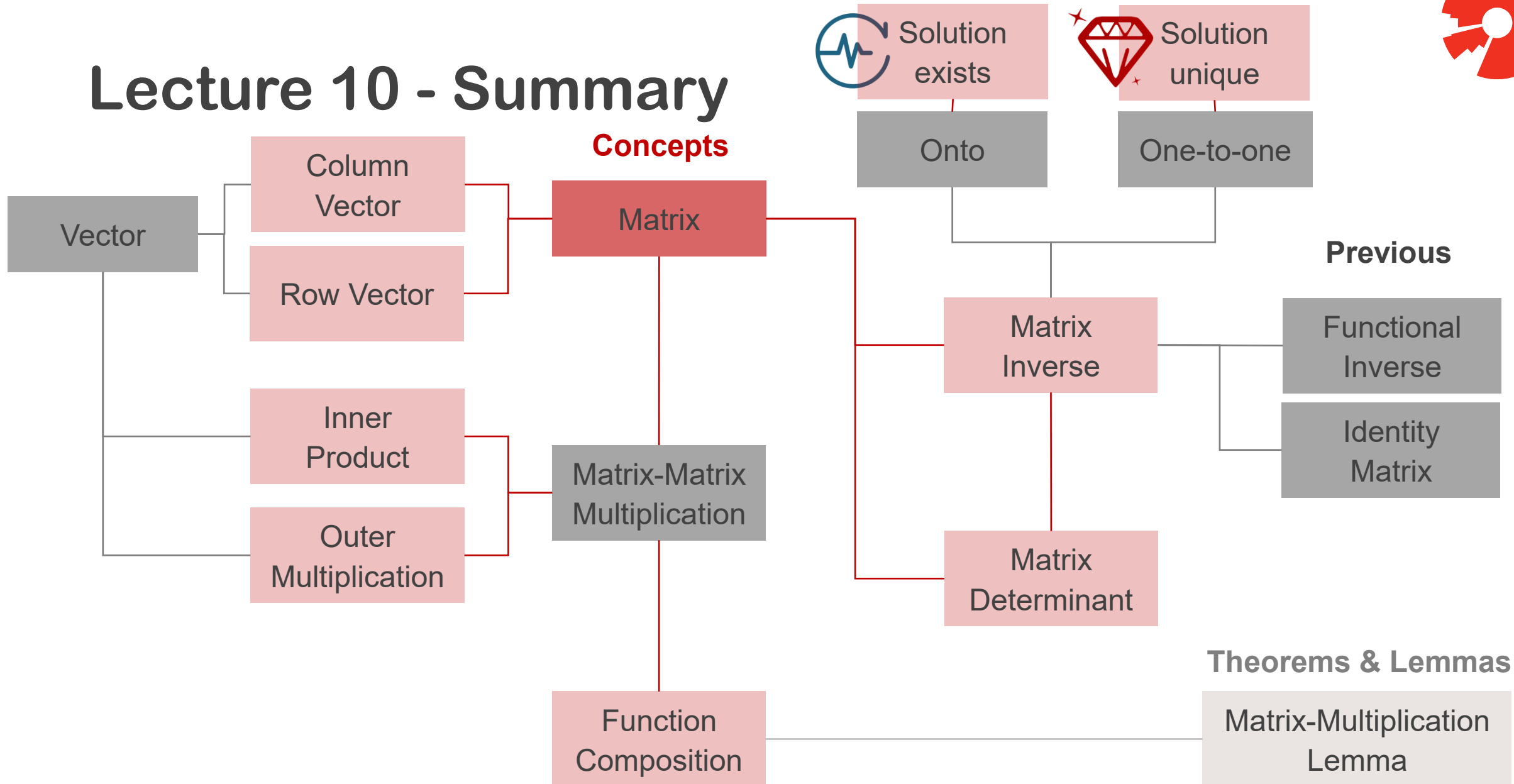
The **matrix determinant** provides one way of knowing if a matrix is invertible

The inverse of a Matrix A is A^{-1} and if A is **invertible**, then $f(x) = A * x$ is an invertible function $\rightarrow$ for $A * x = b$ a solution exists and there is **at most one vector** that can lead to b as it is one-to-one

A solution exists and it is unique



Lecture 10 - Summary





Coordinate Systems (I)

He had a practice of lying in bed in the morning, thinking about mathematics ...

In **1618**, he had an idea... while lying in bed and watching a fly on the ceiling.

He could describe the location of the fly in terms of two numbers: its distance from the two walls.



I think;
therefore
I am.

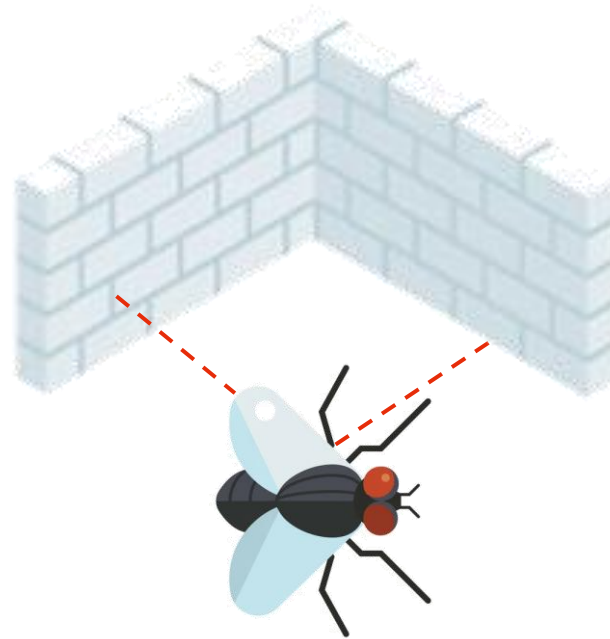
doing linear algebra

Descartes
(1596-1650)



Coordinate Systems (II)

He could describe the location of the fly in terms of two numbers: its distance from the two walls.



I can tell where the fly is based on the walls.
Can you?

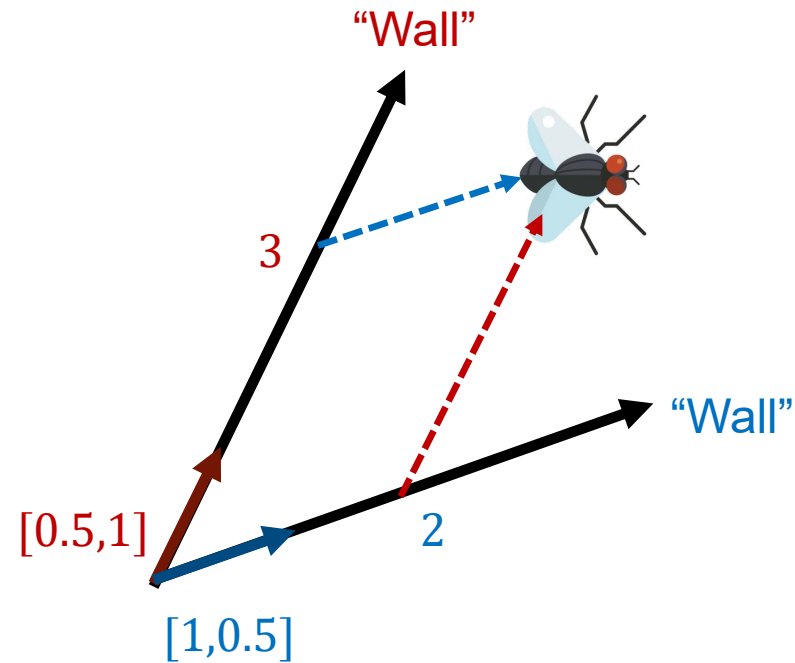
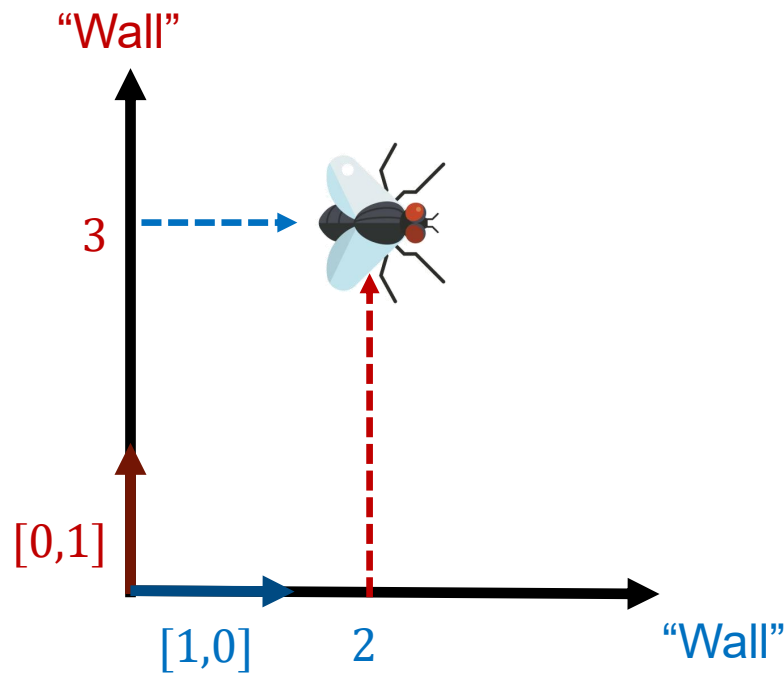




Coordinate Systems (III)

He realized that this works even if the two walls were not perpendicular.

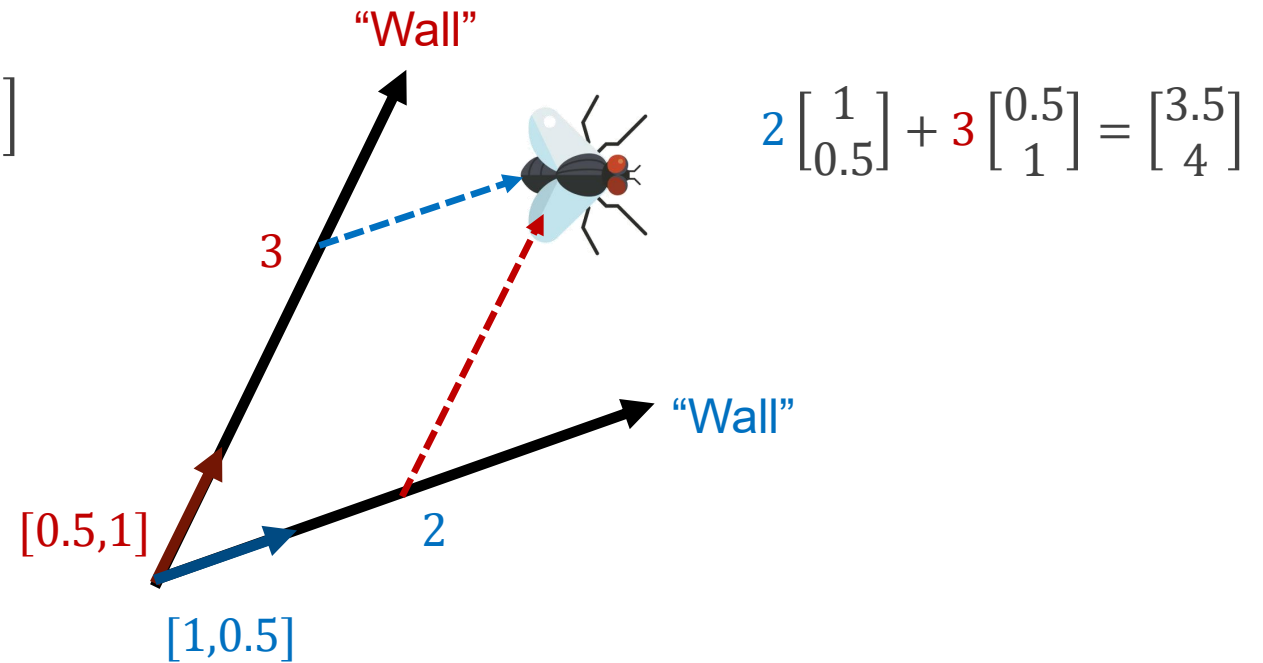
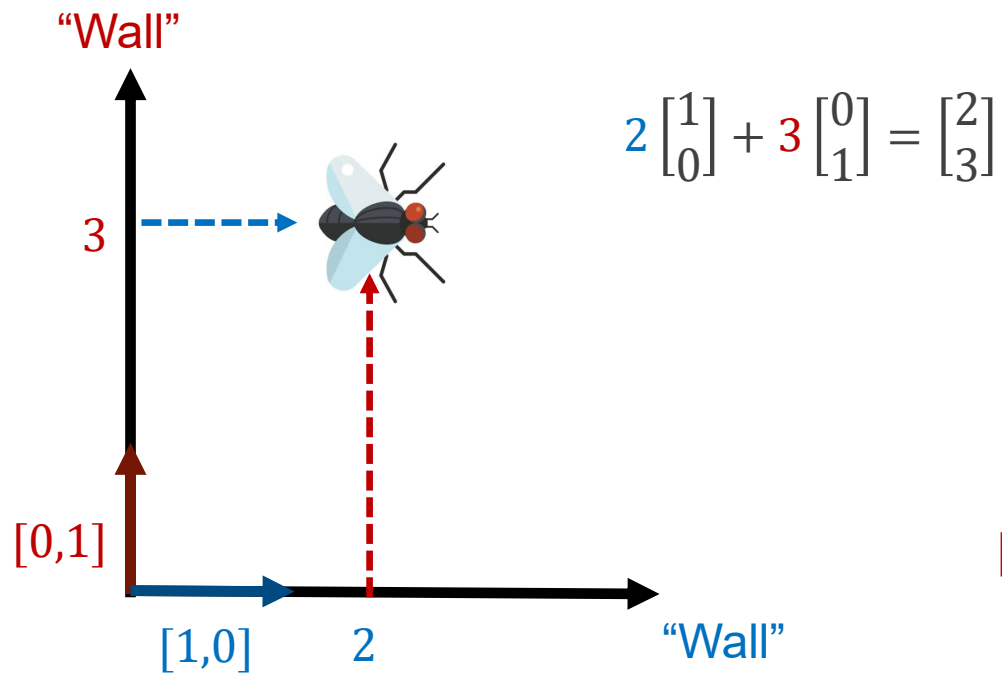
- The walls play role of what we now call *axes*.





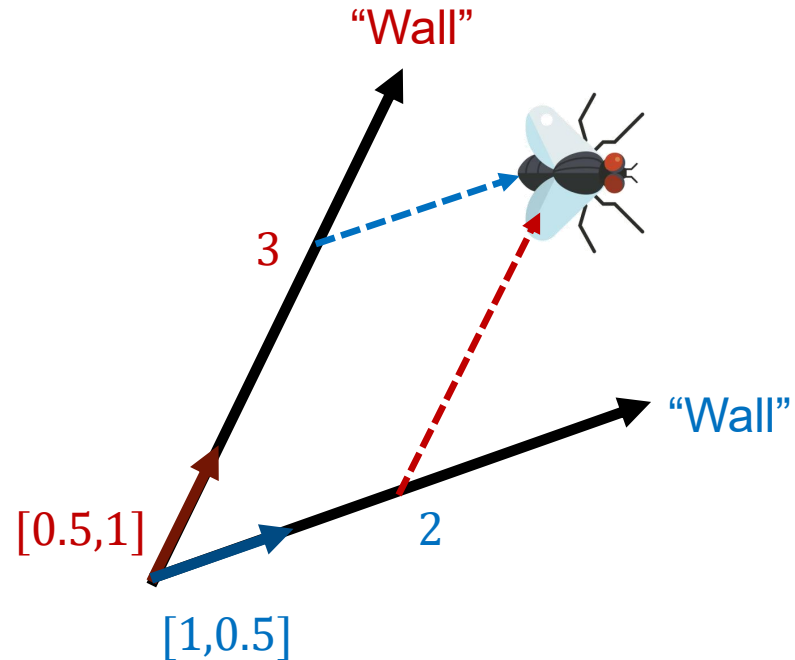
Coordinate Systems (IV)

- ▶ The walls play role of what we now call *axes*.
- ▶ The two numbers are what we now call *coordinates*





Coordinate Systems (V)



Coordinates α_1, α_2

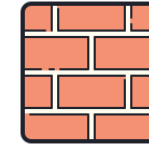
$$2 \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 3 \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix} = v$$

Generators a_1, a_2

Coordinate representation of v in terms of the generators a_1, a_2



Coordinate Systems (VI)



Walls / Axes

In terms of vectors (and generalized beyond two dimensions),

- ▶ *coordinate system* for a vector space $\mathcal{V}$ is specified by **generators** $\mathbf{a}_1, \dots, \mathbf{a}_n$ of $\mathcal{V}$
- ▶ Every vector $\mathbf{v}$ in $\mathcal{V}$ can be written as a linear combination

$$\mathbf{v} = \alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n$$

- ▶ We **represent vector $\mathbf{v}$ by the vector $[\alpha_1, \dots, \alpha_n]$ of coefficients.** called the *coordinate representation* of $\mathbf{v}$ in terms of $\mathbf{a}_1, \dots, \mathbf{a}_n$.

But assigning coordinates to points is not enough. In order to avoid confusion, we must ensure that each point is assigned coordinates in exactly one way. How?

We will discuss unique representation later.

Unique? Sign me up





Coordinate Representation (VII)

Let's start with the easiest coordinate representation → in terms of the standard generators?

Example: What is the coordinate representation of the vector $\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$ in terms of the standard generators?

$$\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\alpha_1 \mathbf{a}_1 \quad \alpha_2 \mathbf{a}_2 \quad \alpha_3 \mathbf{a}_3$

the coordinate representation is $\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$

That shows why we like them – these simple perpendicular walls / axes

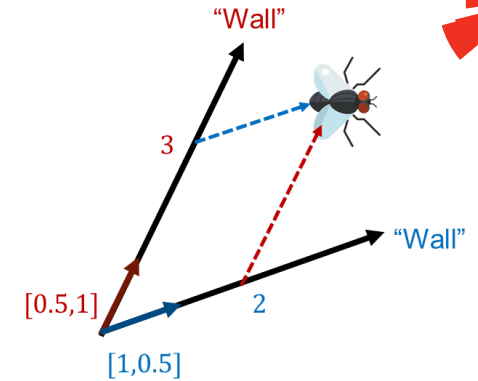


The (standard) vector representation is the same as the coordinate representation using **standard generators**



Coordinate Representation (VIII)

How about when we have generators that are not the standard generators?



Example:

What is the coordinate representation of the vector $v = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$ in terms of $a_1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$?

$$\begin{bmatrix} 3.5 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 3 \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$$

$\alpha_1 \quad a_1 \quad \alpha_2 \quad a_2$

the coordinate representation is $\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Coordinates for the coordinate representation of v using these **two generators (vectors)**

You have a keen eye – but how to find these coordinates?





Coordinate Representation (IX)

Example:

How to find the coordinate representation of the vector $v = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$ in terms of $a_1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$?

$$Au = v$$

$$\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} * [\alpha_1, \alpha_2]^T = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix} \quad \text{Matrix-vector equation}$$

$$\alpha_1 \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix} \quad \text{Linear Combination of columns (generators) of } A$$

$\underbrace{\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}_{a_1} \quad \underbrace{\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}}_{a_2}$

⋮



Coordinate Representation (X)

Example:

How to find the coordinate representation of the vector $v = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$ in terms of $a_1 = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$ and $a_2 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}$?

$$\alpha_1 \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_1 \\ 0.5\alpha_1 \end{bmatrix} + \begin{bmatrix} 0.5\alpha_2 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_1 + 0.5\alpha_2 \\ 0.5\alpha_1 + \alpha_2 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 4 \end{bmatrix}$$

System of LEs

$$\left| \begin{array}{l} \alpha_1 + 0.5\alpha_2 = 3.5 \\ 0.5\alpha_1 + \alpha_2 = 4 \end{array} \right| \xleftrightarrow{R_2 - 0.5R_1} \left| \begin{array}{l} \alpha_1 + 0.5\alpha_2 = 3.5 \\ 0.75\alpha_2 = 2.25 \end{array} \right|$$

(reduce number of unknowns
in second equation)

$$\left| \begin{array}{l} \alpha_1 + 0.5(\mathbf{3}) = 3.5 \\ \alpha_2 = \mathbf{3} \end{array} \right| \xleftrightarrow{\hspace{1cm}} \left| \begin{array}{l} \alpha_1 = 2 \\ \alpha_2 = 3 \end{array} \right|$$

The vector v in the coordinate representation via a_1 and a_2 has the coordinates $u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$



Coordinate Representation (XI)

Let $A = \left[\begin{array}{c|c|c} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{array} \right]$.

Generators (pointing to $\mathbf{a}_1, \dots, \mathbf{a}_n$)

Coefficients (=coordinates) (pointing to $\alpha_1, \alpha_2, \dots, \alpha_n$)

$\mathbf{u} = [\alpha_1, \alpha_2, \dots, \alpha_n]^T$

Coordinate representation of $\mathbf{v}$ (pointing to $\mathbf{u}$)

vector (being represented) (pointing to $\mathbf{v}$)

$\mathbf{v} = A\mathbf{u} = \alpha_1 \mathbf{a}_1 + \cdots + \alpha_n \mathbf{a}_n$

- ▶ “ $\mathbf{u}$ is the coordinate representation of $\mathbf{v}$ in terms of $\mathbf{a}_1, \dots, \mathbf{a}_n$ ” can be written as matrix-vector equation $A\mathbf{u} = \mathbf{v}$

- ▶ To go from a coordinate representation $\mathbf{u}$ to the vector being represented, we multiply A times $\mathbf{u}$.



- ▶ To go from a vector $\mathbf{v}$ to its coordinate representation, we can solve the matrix-vector equation $A\mathbf{u} = \mathbf{v}$.

(Because the columns of A are generators for $\mathcal{V}$ and $\mathbf{v}$ belongs to $\mathcal{V}$, the equation must have at least one solution.)

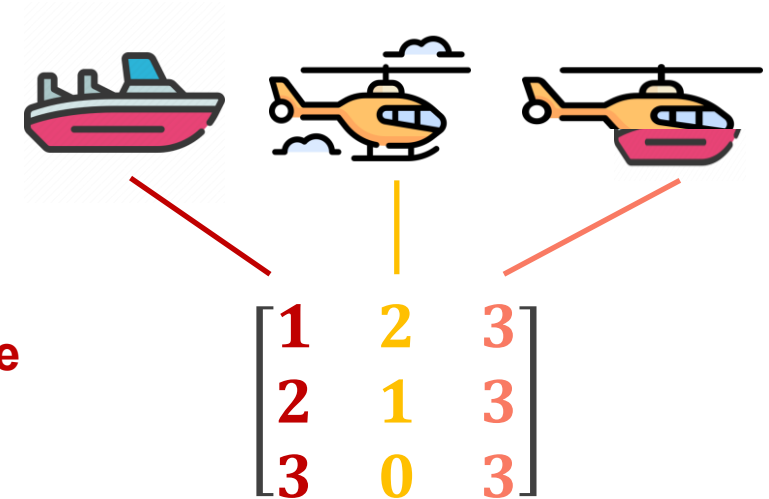
**So we know (=A)
and $\mathbf{v}$, but we
don't know $\mathbf{u}$**



Finding a Set of Generators (I)

Span $\rightarrow$ what we can reach (with the vectors we have)

These vectors span a space. But are they all needed to span the space? Or what is the minimum number to span this space?



Two natural approaches come to mind, the *Grow* algorithm and the *Shrink* algorithm.

We will take some graphs as examples since finding (minimum) spanning forests is actually the same type of problem



Grow Algorithm



Put in, until
you are
spanning

```
def GROW( $\mathcal{V}$ )  
   $S = \emptyset$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $\mathcal{V}$  that is not in  $\text{Span } S$ , and put it in  $S$ .
```

The algorithm stops when there is no vector to add, at which time S spans all of $\mathcal{V}$.

But is it bigger than necessary?



I cannot go anywhere. How
about if I have a helicopter?
How if I add a boat? ...



Shrink Algorithm

```
def SHRINK( $\mathcal{V}$ )  
   $S = \text{some finite set of vectors that spans } \mathcal{V}$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $S$  such that  $\text{Span}(S - \{\mathbf{v}\}) = \mathcal{V}$ , and remove  $\mathbf{v}$  from  $S$ .
```



**Take out but
keep spanning**

The algorithm stops when there is no vector whose removal would leave a spanning set.

The question is, again: is it bigger than necessary?



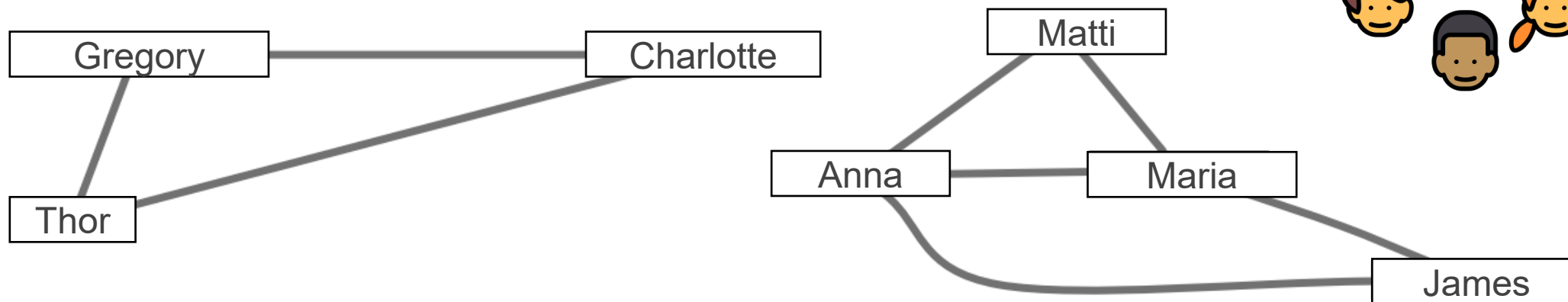
I can go everywhere. Can I still do that if I sell my boaty-copter? (and only keep my boat and helicopter)



Contacts in the
study groups



Minimum Spanning Forest (I)



Definition: A sequence of edges $[\{x_1, x_2\}, \{x_2, x_3\}, \{x_3, x_4\}, \dots, \{x_{k-1}, x_k\}]$ with no repeats is called an x_1 -to- x_k *path*.

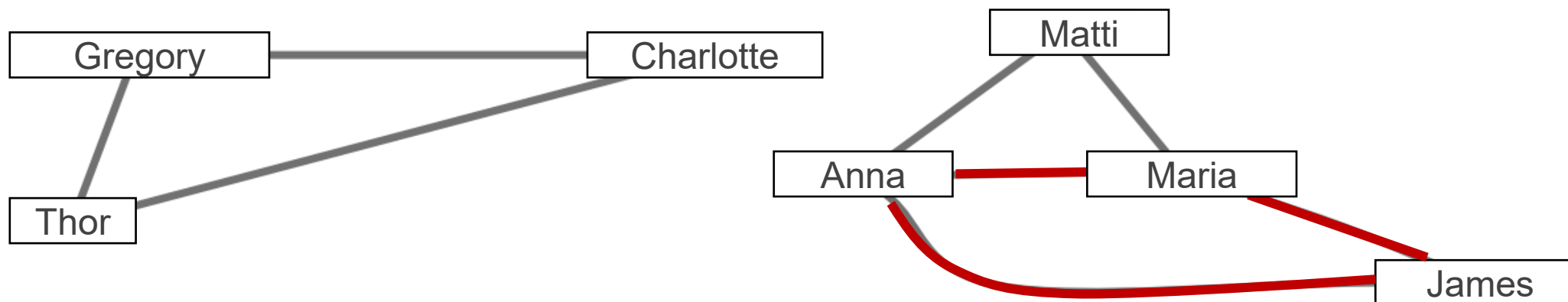
Example Matti -to- James paths in above graph:

- ▶ one goes through Anna
- ▶ one goes through Maria

How can Matti get in touch with James?



Minimum Spanning Forest (II)



Definition: A x -to- x path is called a *cycle*.

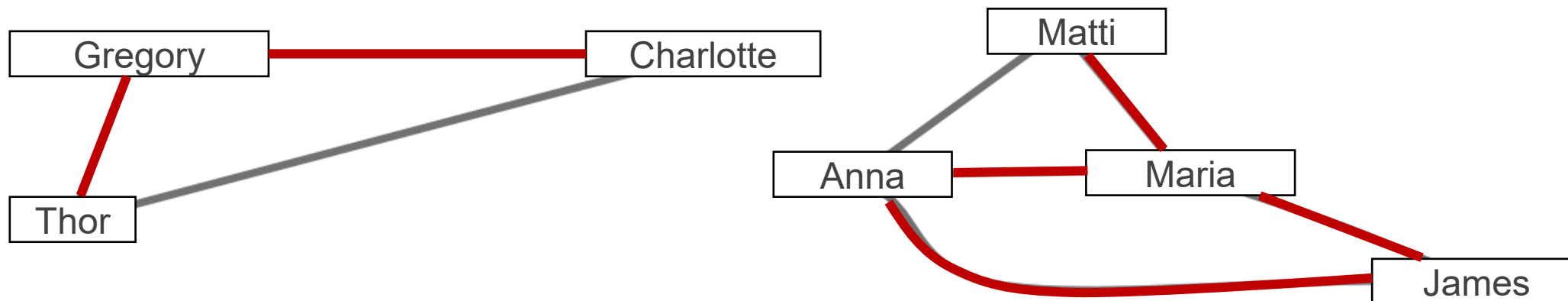
[e.g., James contacts Maria, who contacts Anna, and James gets contacted back from Anna]

Thanks for reaching out, Anna, but I know we meet for studies tomorrow
– I set it up and told Maria!



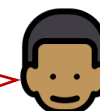


Minimum Spanning Forest (III)



Nice everyone in the two subgroups (components) is **connected** (directly or indirectly) to the other members

Still a **cycle** though



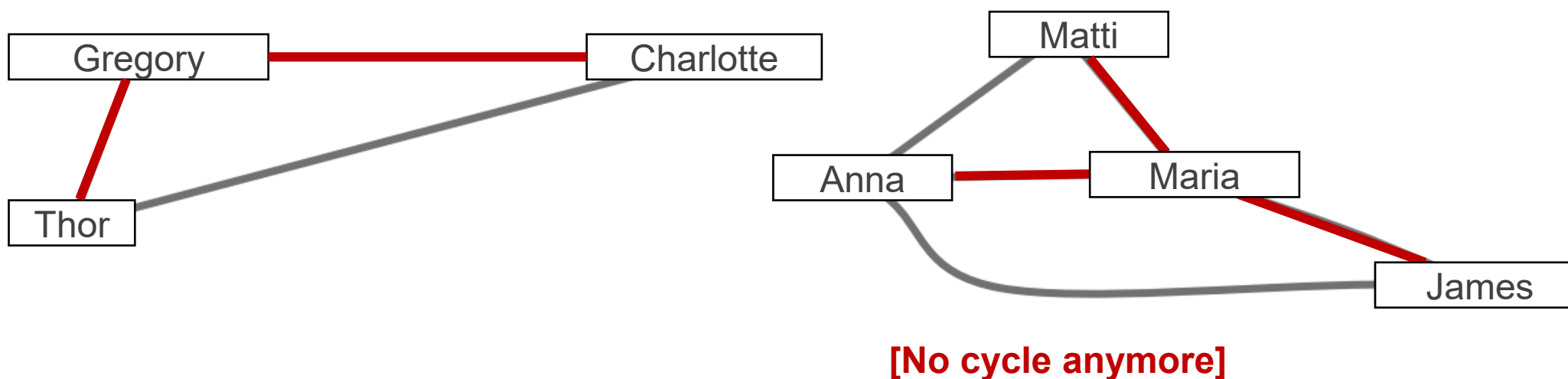
Definition: A set S of edges is *spanning* for a graph G if, for every edge $\{x, y\}$ of G , there is an x -to- y path consisting of edges of S .

Soon we see connection between this use of “spanning” and its use with vectors. 

What can we reach?

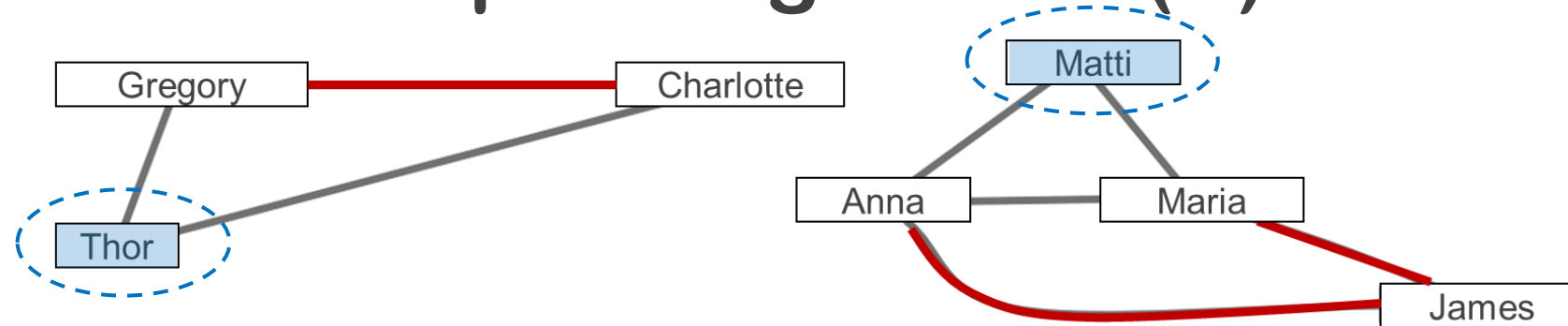


Minimum Spanning Forest (IV)

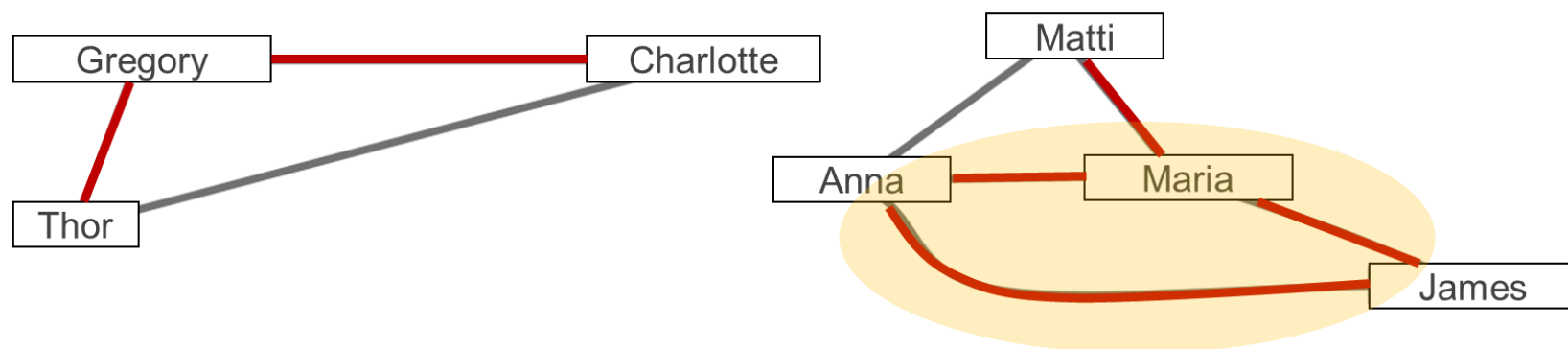


Definition: A set of edges of G is a forest if the set includes no cycles.
(or tree)

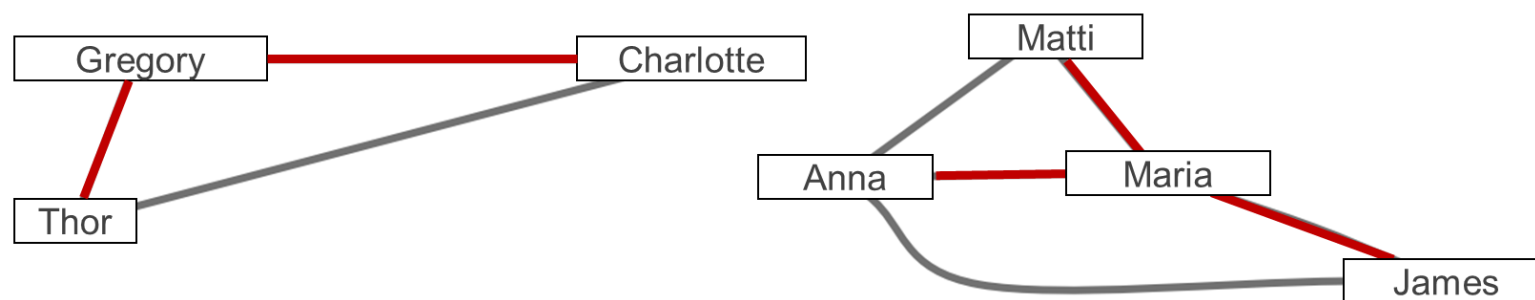
Minimum Spanning Forest (V)



**Not
spanning**



**Spanning
(but Cycle)**

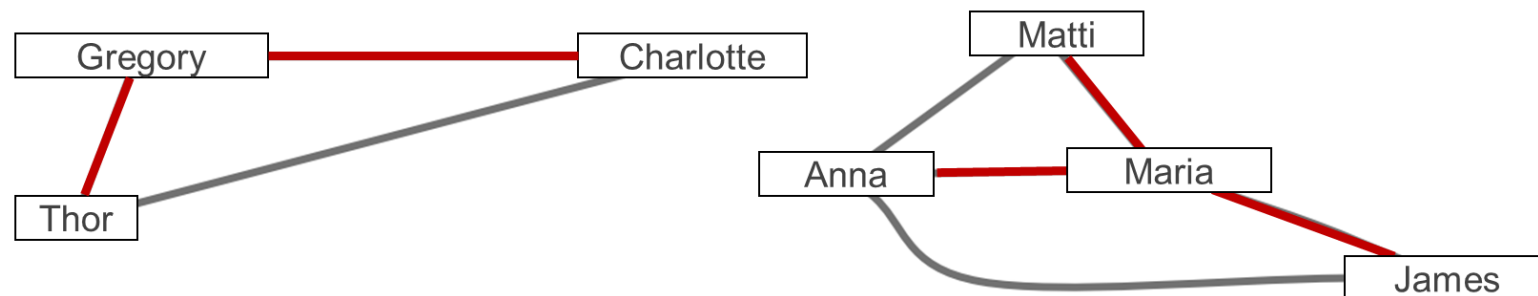


**Spanning
& Forest**





Minimum Spanning Forest (VI)



Spanning
& Forest



Why do we care about this for our **linear algebra** problems?

Spanning → is a set of vectors spanning the vector space (can we reach everywhere in that space)?



Forest → no cycle shows that there only one x-to-y path → no two ways to get to the same “output”



Linear Dependence - Superfluity (I)

Superfluous = “more than is needed or wanted” [Cambridge Dictionary]

Grow and Shrink algorithms both test whether a vector is superfluous in spanning a vector space $\mathcal{V}$. Need a criterion for superfluity.

Superfluous-Vector Lemma: For any set S and any vector $\mathbf{v} \in S$, if $\mathbf{v}$ can be written as a linear combination of the other vectors in S then $\text{Span}(S - \{\mathbf{v}\}) = \text{Span } S$



I see, my boaty-copter is **superfluous** as my boat and helicopter are enough to go to the same places (**span**)



Linear Dependence - Superfluity (II)

Superfluous-Vector Lemma: For any set S and any vector $\mathbf{v} \in S$, if $\mathbf{v}$ can be written as a linear combination of the other vectors in S then $\text{Span}(S - \{\mathbf{v}\}) = \text{Span } S$



Proof: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Suppose $\mathbf{v}_n = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_{n-1} \mathbf{v}_{n-1}$

To show: every vector in $\text{Span } S$ is also in $\text{Span}(S - \{\mathbf{v}_n\})$.

Every vector $\mathbf{v}$ in $\text{Span } S$ can be written as $\mathbf{v} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \dots + \beta_n \mathbf{v}_n$

Substituting for $\mathbf{v}_n$, we obtain

$$\begin{aligned} \mathbf{v} &= \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \dots + \beta_n (\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_{n-1} \mathbf{v}_{n-1}) \\ &= (\beta_1 + \beta_n \alpha_1) \mathbf{v}_1 + (\beta_2 + \beta_n \alpha_2) \mathbf{v}_2 + \dots + (\beta_{n-1} + \beta_n \alpha_{n-1}) \mathbf{v}_{n-1} \end{aligned}$$

which shows that an arbitrary vector in $\text{Span } S$ can be written as a linear combination of vectors in $S - \{\mathbf{v}_n\}$ and is therefore in $\text{Span}(S - \{\mathbf{v}_n\})$.

If we can express $\mathbf{v}_n$ via the other vectors

then we can express any vector in $\text{Span } S$ via the other vectors

So $\mathbf{v}_n$ does not affect the span

QED



Linear Dependence - Superfluity (III)

First Simple Example: Let's assume a set of vectors $S = \{v_1, v_2\}$. Is $\text{span}(S) = \text{span}(S - \{v_2\})$, Meaning is v_2 is in $\text{span}(S - \{v_2\})$ – here simply $\text{span}(\{v_1\})$?

$$v_1 = \begin{bmatrix} -2 \\ 0.5 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0.5 \\ -2 \end{bmatrix}$$

$$v_2 = \alpha_1 v_1$$

$$\begin{bmatrix} 0.5 \\ -2 \end{bmatrix} = \alpha_1 \begin{bmatrix} -2 \\ 0.5 \end{bmatrix}$$

$$\begin{bmatrix} 0.5 \\ -2 \end{bmatrix} = \begin{bmatrix} -2\alpha_1 \\ 0.5\alpha_1 \end{bmatrix}$$

System of LEs

$$\begin{cases} -2\alpha_1 = 0.5 \\ 0.5\alpha_1 = -2 \end{cases}$$



$$\begin{cases} \alpha_1 = -0.25 \\ \alpha_1 = -4 \end{cases}$$

Inconsistent
 $\rightarrow v_2$ is not in $\text{span}(\{v_1\})$



Linear Dependence - Superfluity (IV)

Second Simple Example: Let's assume a set of vectors $S = \{v_1, v_2\}$. Is $\text{span}(S) = \text{span}(S - \{v_2\})$, Meaning is v_2 is in $\text{span}(S - \{v_2\})$ – here simply $\text{span}(\{v_1\})$?

$$v_1 = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$v_2 = \alpha_1 v_1$$

$$\begin{bmatrix} 2 \\ -2 \end{bmatrix} = \alpha_1 \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -2\alpha_1 \\ 2\alpha_1 \end{bmatrix}$$

System of LEs

$$\left| \begin{array}{c} -2\alpha_1 = 2 \\ 2\alpha_1 = -2 \end{array} \right| \longleftrightarrow \left| \begin{array}{c} \alpha_1 = -1 \\ \alpha_1 = -1 \end{array} \right|$$

Consistent

→ v_2 is in $\text{span}(\{v_1\})$

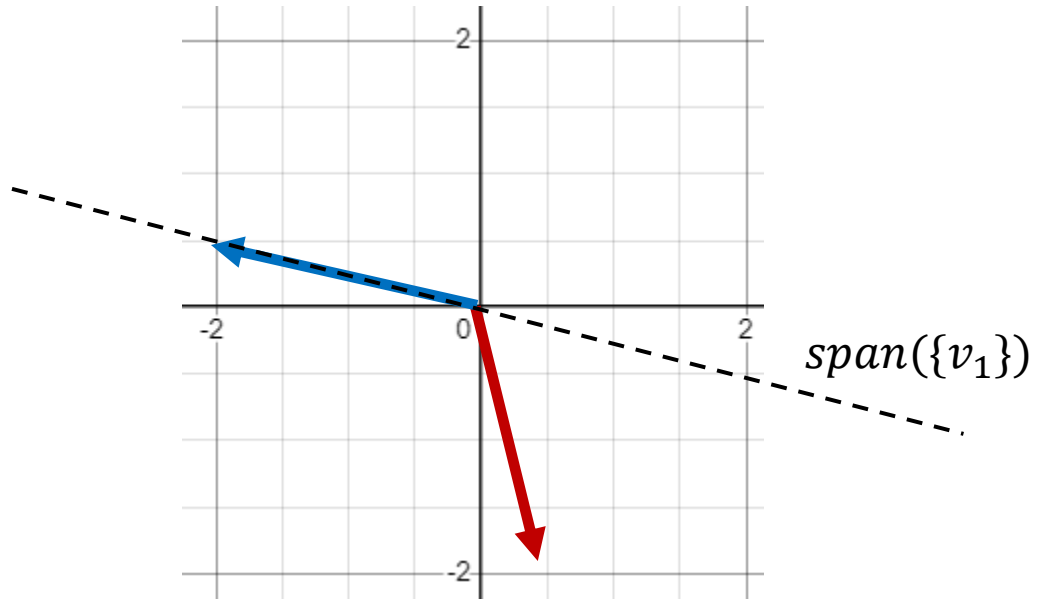
As $-1 * v_1 = v_2$



Linear Dependence - Superfluity (V)

$$v_1 = \begin{bmatrix} -2 \\ 0.5 \end{bmatrix}$$

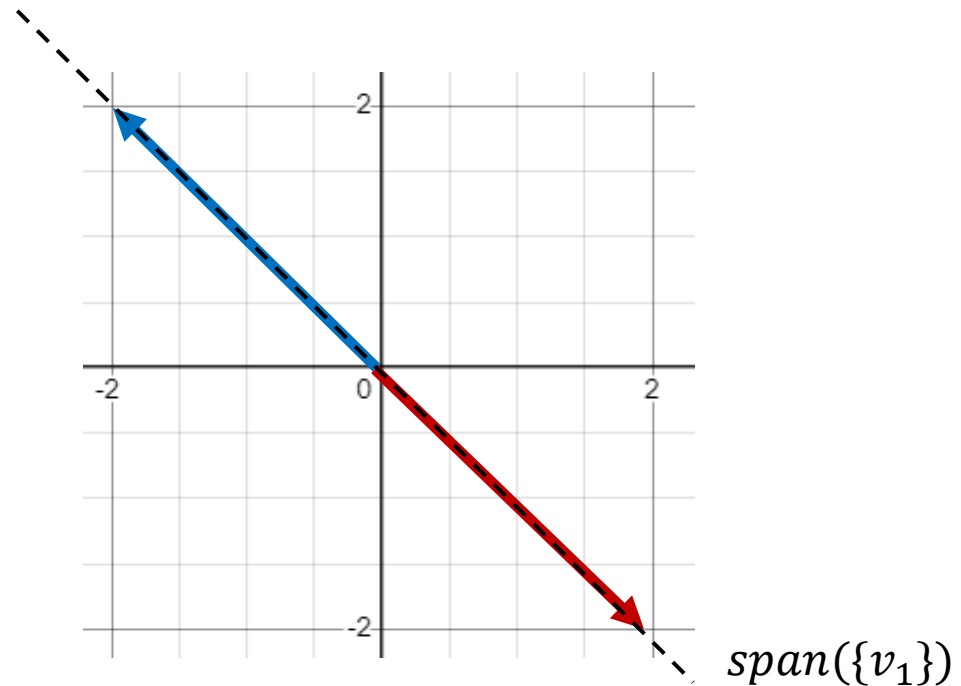
$$v_2 = \begin{bmatrix} 0.5 \\ -2 \end{bmatrix} \rightarrow v_2 \text{ is not in } \text{span}(\{v_1\})$$



$$v_1 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix} \rightarrow v_2 \text{ is in } \text{span}(\{v_1\})$$

As $-1 * v_1 = v_2$



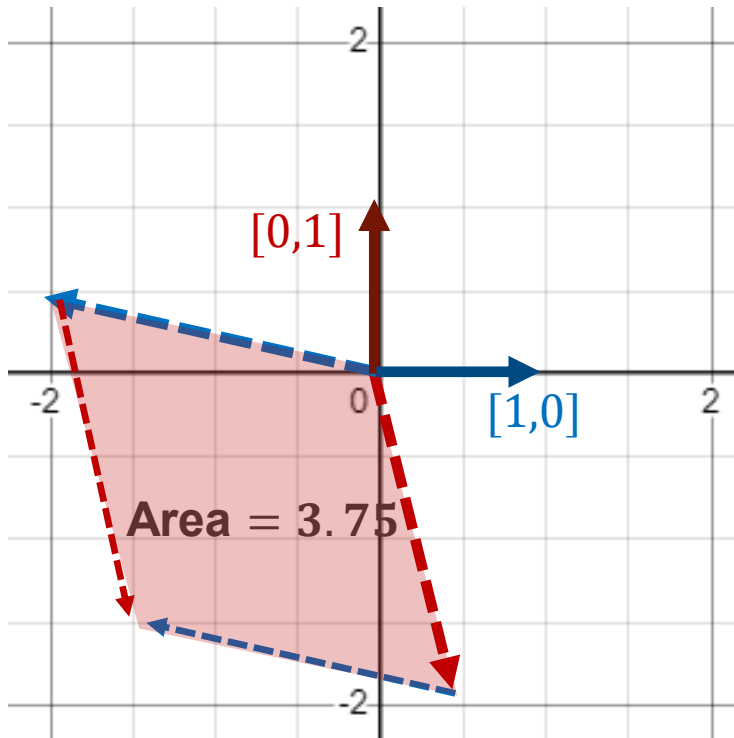


$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

Superfluity – Link to Determinants

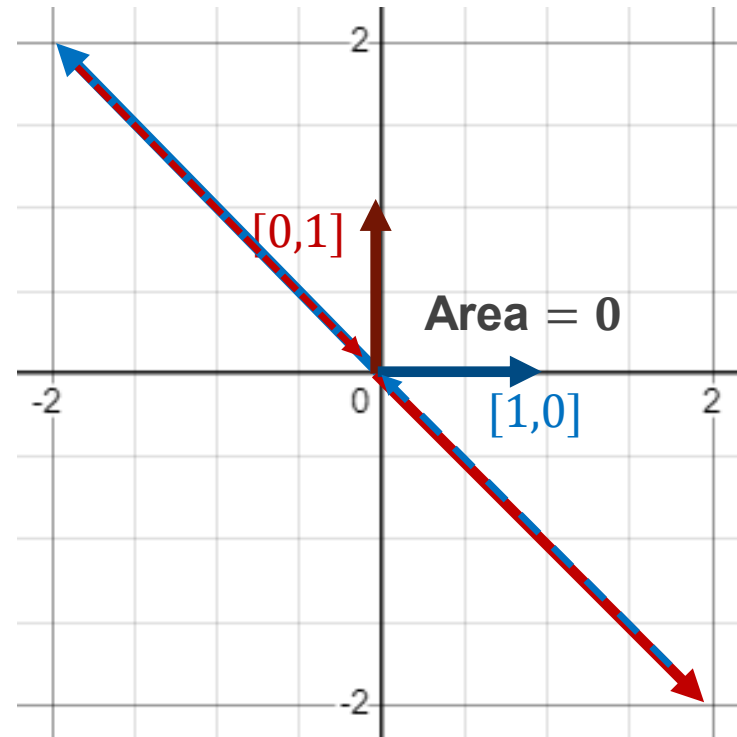
$$A = \begin{bmatrix} -2 & 0.5 \\ 0.5 & -2 \end{bmatrix}$$

$$\det A = 4 - 0.25 = 3.75$$



$$A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$\det A = 4 - 4 = 0$$



Note: the **span** of the vectors is the same

→ Second column vector is **superfluous** to the span

→ ... (in a couple of slides) ...



Linear Dependence (I)

Definition: Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are *linearly dependent* if the zero vector can be written as a **nontrivial** linear combination of the vectors:

Not just $\alpha_1, \alpha_2, \dots, \alpha_n = 0$
[trivial]

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$$

In this case, we refer to the linear combination as a *linear dependency* in $\mathbf{v}_1, \dots, \mathbf{v}_n$.

On the other hand, if the *only* linear combination that equals the zero vector is the trivial linear combination, we say $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly *independent*.

Simply speaking, if we can have nonzero α coefficients so that one or multiple vectors can „compensate“ each others effects





Linear Dependence (II)

First Simple Example: Are the vectors $v_1 = \begin{bmatrix} -2 \\ 0.5 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0.5 \\ -2 \end{bmatrix}$ linearly independent?

$$0 = \alpha_1 v_1 + \alpha_2 v_2$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} -2 \\ 0.5 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0.5 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} -2\alpha_1 \\ 0.5\alpha_1 \end{bmatrix} + \begin{bmatrix} 0.5\alpha_2 \\ -2\alpha_2 \end{bmatrix}$$

$$= \begin{bmatrix} -2\alpha_1 + 0.5\alpha_2 \\ 0.5\alpha_1 - 2\alpha_2 \end{bmatrix}$$

System of **homog. LEs** 😲

$$\begin{vmatrix} -2\alpha_1 + 0.5\alpha_2 = 0 \\ 0.5\alpha_1 - 2\alpha_2 = 0 \end{vmatrix} \xleftrightarrow{R_2 + 4R_1} \begin{vmatrix} -2\alpha_1 + 0.5\alpha_2 = 0 \\ -7.5\alpha_1 = 0 \end{vmatrix}$$

(reduce number of unknowns
in second equation)

$$\begin{vmatrix} -2(0) + 0.5\alpha_2 = 0 \\ \alpha_1 = 0 \end{vmatrix} \xleftrightarrow{\quad} \begin{vmatrix} \alpha_2 = 0 \\ \alpha_1 = 0 \end{vmatrix} \rightarrow \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Only $0v_1 + 0v_2 = 0$

(i.e., zero vector can only be written via the trivial linear combination) $\rightarrow v_1$ and v_2 are **linearly independent**



Linear Dependence (III)

First Simple Example: Are the vectors $v_1 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$ linearly independent?

$$0 = \alpha_1 v_1 + \alpha_2 v_2$$

$$\begin{aligned} \begin{bmatrix} 0 \\ 0 \end{bmatrix} &= \alpha_1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} -2\alpha_1 \\ 2\alpha_1 \end{bmatrix} + \begin{bmatrix} 2\alpha_2 \\ -2\alpha_2 \end{bmatrix} \\ &= \begin{bmatrix} -2\alpha_1 + 2\alpha_2 \\ 2\alpha_1 - 2\alpha_2 \end{bmatrix} \end{aligned}$$

System of **homog. LEs**

$$\left| \begin{array}{l} -2\alpha_1 + 2\alpha_2 = 0 \\ 2\alpha_1 - 2\alpha_2 = 0 \end{array} \right| \xleftrightarrow{R_2 + R_1} \left| \begin{array}{l} -2\alpha_1 + 2\alpha_2 = 0 \\ 0 = 0 \end{array} \right|$$

(reduce number of unknowns
in second equation)

$$\left| \begin{array}{l} \alpha_2 = \alpha_1 \\ 0 = 0 \end{array} \right| \Rightarrow \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\alpha_1 v_1 + \alpha_2 v_2 = 0$ for any linear combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\rightarrow v_1$ and v_2 are **linearly dependent** (not linearly independent)



Linear Dependence (IV)

$$v_1 = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\alpha_1 v_1 + \alpha_2 v_2 = 0$ for any linear combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ including

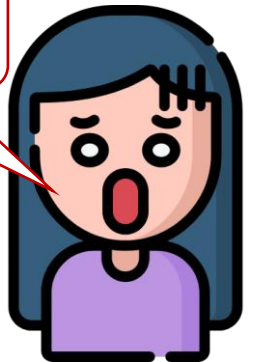
$$0 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$1 \begin{bmatrix} -2 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5 \begin{bmatrix} -2 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

...

They “cancel”
each other out...



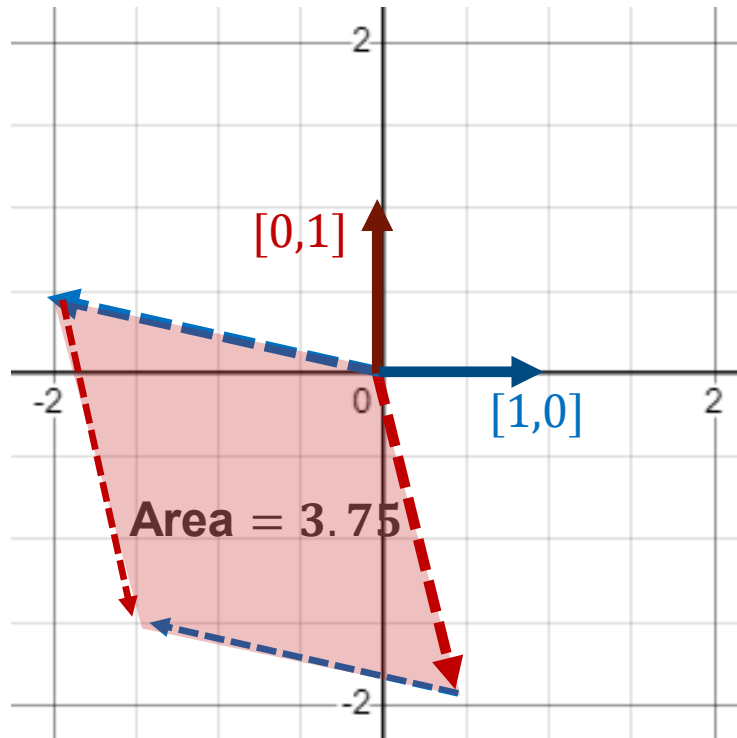


$$\det A = |A| = a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$$

Linear Dependence – Link to Determinants

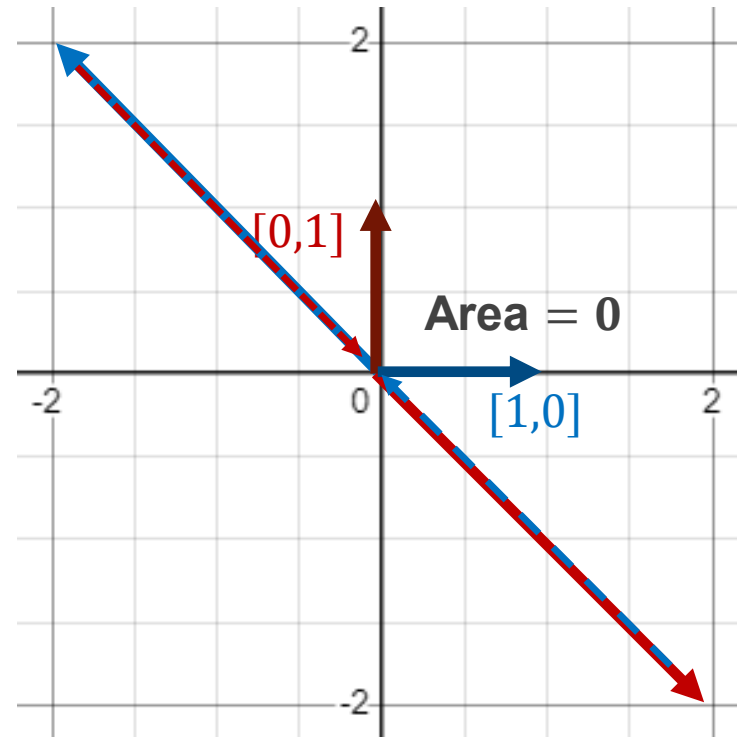
$$A = \begin{bmatrix} -2 & 0.5 \\ 0.5 & -2 \end{bmatrix}$$

$$\det A = 4 - 0.25 = 3.75$$



$$A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$\det A = 4 - 4 = 0$$



Note: the **span** of the vectors is the same

→ Second column vector is **superfluous** to the span

→ The column vectors are **linearly dependent**



Linear Dependence – Not just 2D (I)

3D Example: Are the vectors in v linearly independent?

$$\begin{bmatrix} \textcolor{red}{1} & \textcolor{yellow}{2} & \textcolor{red}{3} \\ \textcolor{red}{2} & \textcolor{yellow}{1} & \textcolor{red}{3} \\ \textcolor{red}{3} & \textcolor{yellow}{0} & \textcolor{red}{3} \end{bmatrix}$$





$$\mathbf{0} = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 \\ 3\alpha_1 + 3\alpha_3 \end{bmatrix}$$

System of **homog. LEs**

$$\begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \\ 2\alpha_1 + \alpha_2 + 3\alpha_3 = 0 \\ 3\alpha_1 + 3\alpha_3 = 0 \end{cases}$$

(reduce number of unknowns
in second equation)

$R_2 - 0.5R_1$



$$\begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \\ 1.5\alpha_1 + 1.5\alpha_3 = 0 \\ 3\alpha_1 + 3\alpha_3 = 0 \end{cases}$$

$R_3 - 2R_2$

$$\begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \\ 1.5\alpha_1 + 1.5\alpha_3 = 0 \\ 0 = 0 \end{cases}$$



$$\begin{cases} \alpha_1 + 2\alpha_2 + 3(-\alpha_1) = 0 \\ \alpha_3 = -\alpha_1 \\ 0 = 0 \end{cases}$$

$$\begin{cases} -2\alpha_1 + 2\alpha_2 = 0 \\ \alpha_3 = -\alpha_1 \end{cases}$$



$$\begin{cases} \alpha_2 = \alpha_1 \\ \alpha_3 = -\alpha_1 \end{cases}$$



Linear Dependence – Not just 2D (II)

3D Example: Are the vectors in v linearly independent?

$$\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{2} & \mathbf{1} & \mathbf{3} \\ \mathbf{3} & \mathbf{0} & \mathbf{3} \end{bmatrix}$$





$$\begin{cases} \alpha_2 = \alpha_1 \\ \alpha_3 = -\alpha_1 \end{cases} \quad \rightarrow \quad \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ -\alpha_1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$ for any linear combination of $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

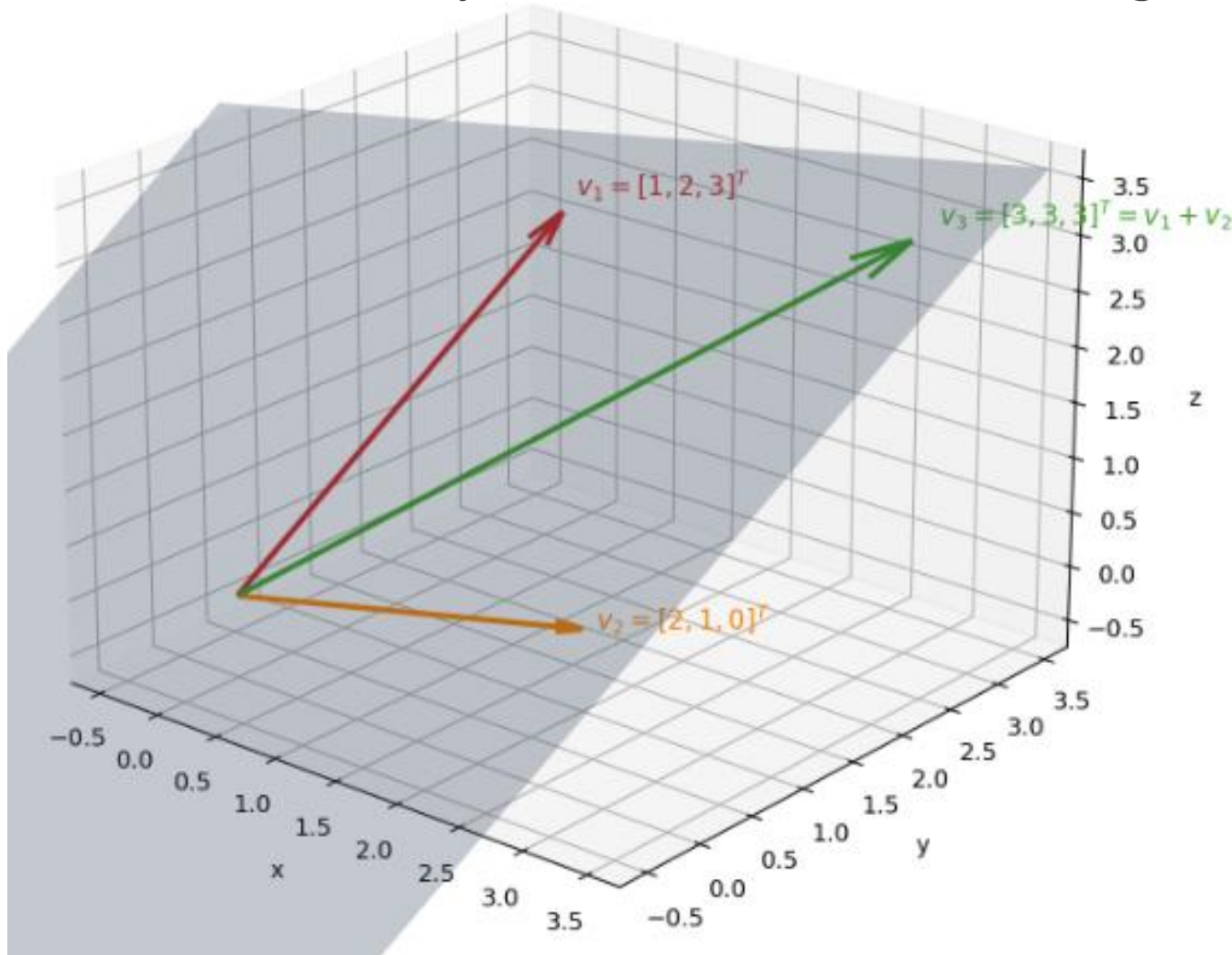
→ The three vectors are **linearly dependent**
(not linearly independent)

So, my boat, helicopter and boatcopter are **linearly dependent**?
(At least) one is **superfluous**?!





Linear Dependence – Not just 2D (III)



$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 0 & 3 \end{bmatrix}$$





v_3 is in the hyperplane spanned by v_1 and v_2 , so it is in their span → **superfluous**

However, since the three vectors are linearly dependent, there are also other subsets that span the same subspace – her: v_1 and v_2 , v_1 and v_3 , v_2 and v_3



Linear (In)Dependence - Properties (I)

Other vectors
„compensate“ for it

Linear-Dependence Lemma Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be vectors.
A vector $\mathbf{v}_i$ is in the span of the other vectors
if and only if
the zero vector can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$
in which the coefficient of $\mathbf{v}_i$ is nonzero.

In graphs, the Linear-Dependence Lemma states that an edge e is in the span of other edges if there is a cycle consisting of e and a subset of the other edges.





Linear (In)Dependence - Properties (II)



Proof: *First direction:* Suppose $\mathbf{v}_i$ is in the span of the other vectors. That is, there exist coefficients $\alpha_1, \dots, \alpha_{n-1}$ such that

$$\mathbf{v}_i = \alpha_1 \mathbf{v}_1 + \dots + \alpha_{i-1} \mathbf{v}_{i-1} + \alpha_{i+1} \mathbf{v}_{i+1} + \dots + \alpha_n \mathbf{v}_n$$

Moving $\mathbf{v}_i$ to the other side, we can write

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + (-1)\mathbf{v}_i + \dots + \alpha_n \mathbf{v}_n$$

which shows that the all-zero vector can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$ in which the coefficient of $\mathbf{v}_i$ is nonzero.



Linear (In)Dependence - Properties (III)



Proof: Now for the other direction. Suppose there are coefficients $\alpha_1, \dots, \alpha_n$ such that

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_i \mathbf{v}_i + \dots + \alpha_n \mathbf{v}_n$$

and such that $\alpha_i \neq 0$. **nonzero**

Dividing both sides by α_i yields

$$\mathbf{0} = (\alpha_1/\alpha_i) \mathbf{v}_1 + (\alpha_2/\alpha_i) \mathbf{v}_2 + \dots + \mathbf{v}_i + \dots + (\alpha_n/\alpha_i) \mathbf{v}_n$$

Moving every term from right to left except $\mathbf{v}_i$ yields

$$-(\alpha_1/\alpha_i) \mathbf{v}_1 - (\alpha_2/\alpha_i) \mathbf{v}_2 - \dots - (\alpha_n/\alpha_i) \mathbf{v}_n = \mathbf{v}_i$$

**v_i can be represented
in terms of $v_1, \dots, v_n$**

QED



Linear (In)Dependence - Properties (IV)

Remember (some slides ago)

Linear-Dependence Lemma Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be vectors. A vector $\mathbf{v}_i$ is in the span of the other vectors if and only if the zero vector can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$ in which the coefficient of $\mathbf{v}_i$ is nonzero.

Contrapositive:

$\mathbf{v}_i$ is not in the space of the other vectors

if and only if

for any linear combination equaling the zero vector

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_i \mathbf{v}_i + \dots + \alpha_n \mathbf{v}_n$$

it must be that the coefficient α_i is zero.



Exercise: Linear Dependence

You are given the following vectors:

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- 1) Show whether u is in $\text{Span}(\{v_1, v_2\})$.
- 2) Are v_1, v_2 and v_3 linearly independent? Show work!
- 3) Assume that the four vectors v_1, v_2, v_3, u form a set of vectors S . Show that $\text{Span}(S)$ is equal to $\text{Span}(S - \{u\})$.
- 4) How is the result in (3) linked to the concept of superfluous vectors? Is u superfluous to $\text{Span}(S)$?



Solution

Exercise: Linear Dependence

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- a) To show that vector u is in $\text{Span}(\{v_1, v_2\})$ we need to show that there is a linear combination of v_1 and v_2 that is u (can “reach” u). This means, there are alpha coefficients, such that $u = \alpha_1 v_1 + \alpha_2 v_2$.

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha_2 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 + \alpha_2 \\ \alpha_2 \end{bmatrix}$$

Set of linear equations to solve

$$\begin{array}{l} \alpha_1 = 1 \\ \alpha_1 + \alpha_2 = 1 \\ \alpha_2 = 1 \end{array} \quad \longleftrightarrow \quad \begin{array}{l} \alpha_1 = 1 \\ 1 + 1 = 1 \\ \alpha_2 = 1 \end{array}$$

The second equation does **not** hold. Thus, there is no linear combination of v_1 and v_2 that equals u . We can conclude that u is **not** in $\text{Span}(\{v_1, v_2\})$



Solution

Exercise: Linear Dependence

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- b) To show that v_1 , v_2 , and v_3 are linearly independent, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$\mathbf{0} = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \quad \text{only with} \quad \alpha_1, \alpha_2, \alpha_3 = 0$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha_2 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha_3 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_2 \end{bmatrix}$$

Set of linear equations to solve

$$\begin{array}{c} \alpha_1 = 0 \\ \alpha_1 + \alpha_2 + \alpha_3 = 0 \\ \alpha_2 = 0 \end{array} \quad \longleftrightarrow \quad \begin{array}{c} \alpha_1 = 0 \\ 0 + 0 + \alpha_3 = 0 \\ \alpha_2 = 0 \end{array} \quad \longleftrightarrow \quad \begin{array}{c} \alpha_1 = 0 \\ \alpha_3 = 0 \\ \alpha_2 = 0 \end{array}$$

Thus, the only linear combination of v_1, v_2, v_3 for the zero vector is the trivial one ($\alpha_1 = 0, \alpha_2 = 0, \alpha_3 = 0$). We can conclude that v_1, v_2, v_3 are **linearly independent**.



Solution

Exercise: Linear Dependence

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- c) To show that $\text{Span}(S)$ is equal to $\text{Span}(S - \{u\})$, meaning that vector u is superfluous, one can simply show that u is in $\text{Span}(\{v_1, v_2, v_3\})$ which is the same as $\text{Span}(S - \{u\})$.

In general, vector u is in the span of vectors v_1, v_2, v_3 if there is a linear combination of these vectors for it, meaning there are alpha coefficients, such that $u = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$.

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha_2 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \alpha_3 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_1 + \alpha_2 + \alpha_3 \\ \alpha_2 \end{bmatrix}$$

Set of linear equations to solve

$$\begin{array}{l|l|l} \begin{array}{l} \alpha_1 = 1 \\ \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha_2 = 1 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 = 1 \\ 1 + 1 + \alpha_3 = 1 \\ \alpha_2 = 1 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 = 1 \\ \alpha_3 = -1 \\ \alpha_2 = 1 \end{array} \end{array}$$

Thus, there is a linear combination of v_1, v_2, v_3 with $\alpha_1 = 1$, $\alpha_2 = 1$, and $\alpha_3 = -1$ for u . Thus, $\text{Span}(S) = \text{Span}(S - \{u\})$ meaning that u is superfluous.

**Solution****Exercise: Linear Dependence**

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

- d) **Superfluous-Vector Lemma:** For any set S and any vector $\mathbf{v} \in S$, if $\mathbf{v}$ can be written as a linear combination of the other vectors in S then $\text{Span}(S - \{\mathbf{v}\}) = \text{Span } S$

Since the results from the previous task show that there is a linear combination of v_1, v_2, v_3 with $\alpha_1 = 1$, $\alpha_2 = 1$, and $\alpha_3 = -1$ for u , we can conclude that u is superfluous (for $\text{Span}(S)$).



Some Key Take-aways (Lecture 11)

A **Minimum Spanning Forest (MSF)** used in graph theory can be used to illustrate spanning. A forest has no cycles and spanning means there is an x-to-y path for edges

The **Growth algorithm** (adds vectors not in span) and the **Shrink algorithm** (removes vectors while maintaining span) can obtain a generating set for a vector space

The **superfluous-vector lemma** states that if a vector in a set of vectors can be represented by a linear combination of the other vectors, then its removal does not affect the span

The **linear-dependence lemma** states that a vector v_i is in the span of other vectors [and, thus, superfluous for the span] iff the zero vector can be written as a linear combination of the other vectors with v_i 's coefficient being nonzero

Linear
Combinations
are essential
for this



Lecture 11 - Summary

Concepts

Basis

Theorems & Lemmas

Linear-Dependence
Lemma

Superfluous-Vector
Lemma

Span

Linear
Combination

Previous

Vector Space

Coefficients

Coordinate
System

Coordinate
Representation

Minimum
Spanning Forest

Tools

Grow
Algorithm

Shrink
Algorithm

Linear
Independence





Thank You For Your Attention!



**Accomplishment unlocked:
Lecture 11 Completed**



APPENDIX



(Note: row vectors used here – still same logic)

Coordinate Representation (II)

Example: What is the coordinate representation of the vector $[6, 3, 2, 5]$ in terms of the vectors $[2, 2, 2, 3]$, $[1, 0, -1, 0]$, $[0, 1, 0, 1]$?

Since

$$[6, 3, 2, 5] = \overset{\alpha_1}{2} \overset{a_1}{[2, 2, 2, 3]} + \overset{\alpha_2}{2} \overset{a_2}{[1, 0, -1, 0]} - \overset{\alpha_3}{1} \overset{a_3}{[0, 1, 0, 1]},$$

the coordinate representation is $\underset{\alpha_1 \alpha_2 \alpha_3}{[2, 2, -1]}$.

Example: What is the coordinate representation of the vector $[0, 0, 0, 1]$ in terms of the vectors $[1, 1, 0, 1]$, $[0, 1, 0, 1]$, and $[1, 1, 0, 0]$?

Since

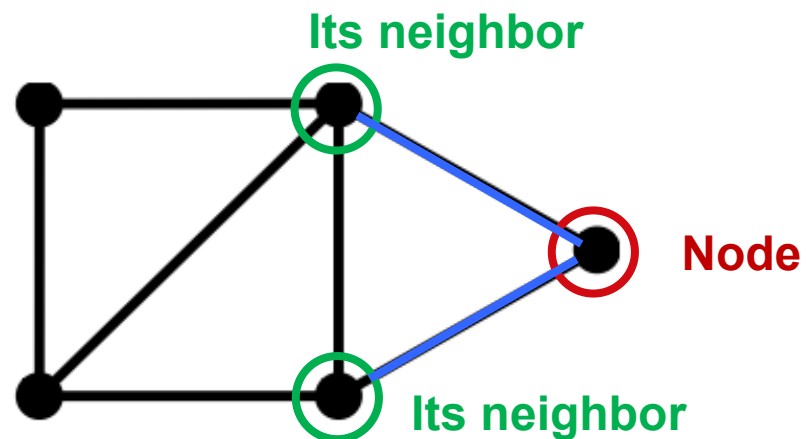
$$[0, 0, 0, 1] = \overset{\alpha_1}{1} \overset{a_1}{[1, 1, 0, 1]} + \overset{\alpha_2}{0} \overset{a_2}{[0, 1, 0, 1]} - \overset{\alpha_3}{1} \overset{a_3}{[1, 1, 0, 0]}$$

the coordinate representation of $[0, 0, 0, 1]$ is $\underset{\alpha_1 \alpha_2 \alpha_3}{[1, 0, -1]}$.



When Greed fails (I)

Is it obvious that Grow algorithm and Shrink algorithm find smallest sets of generators? Look at example for a problem in *graphs*...



Points are called *nodes*, links are called *edges*.

Each edge has two *endpoints*, the nodes it connects. The endpoints of an edge are *neighbors*.





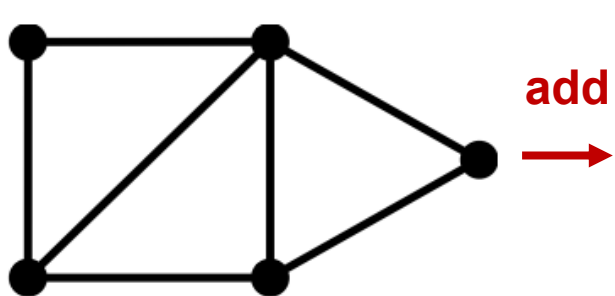
When Greed fails (II)

Definition: A dominating set in a graph is a set S of nodes such that every node is in S or a neighbor of a node in S .

We have different choices which ones to add – ONE example below

Grow Algorithm:
initialize $S = \emptyset$
while S is not a dominating set,
add a node to S that is not currently adjacent to S

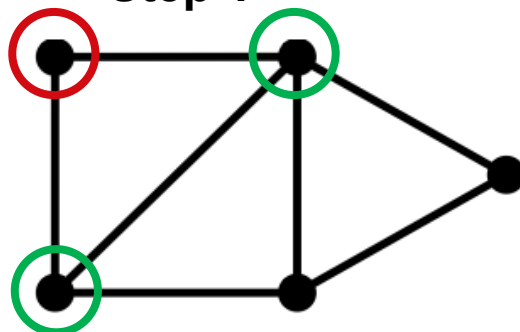
Empty Set



add



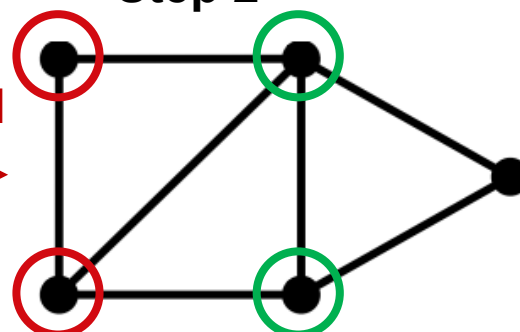
Step 1



add



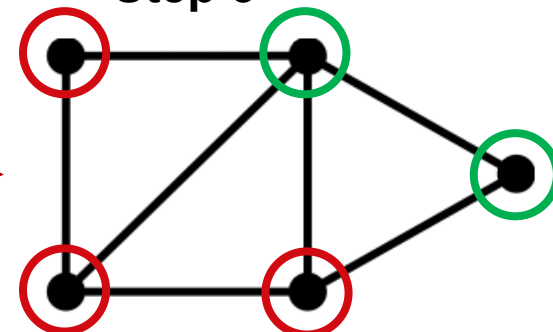
Step 2



add



Step 3

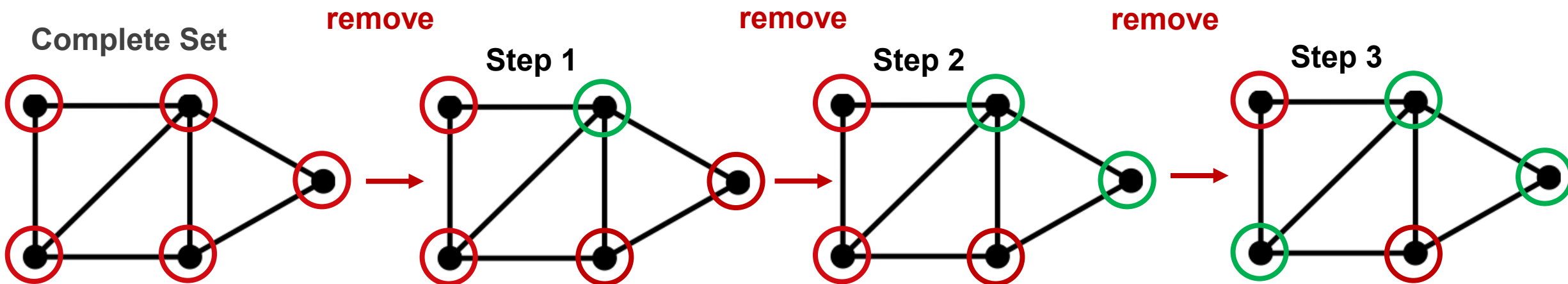




When Greed fails (III)

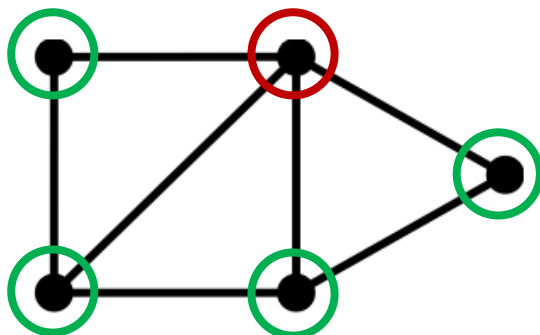
We have different choices which nodes to remove in each step – ONE example below

Shrink Algorithm:
 initialize $S = \text{all nodes}$
 while there is a node x such that $S - \{x\}$ is a dominating set,
 remove x from S



When Greed fails (IV)

Optimal Solution (dominating with minimum number of nodes)



We may end up with the optimal solution, but there is no guarantee



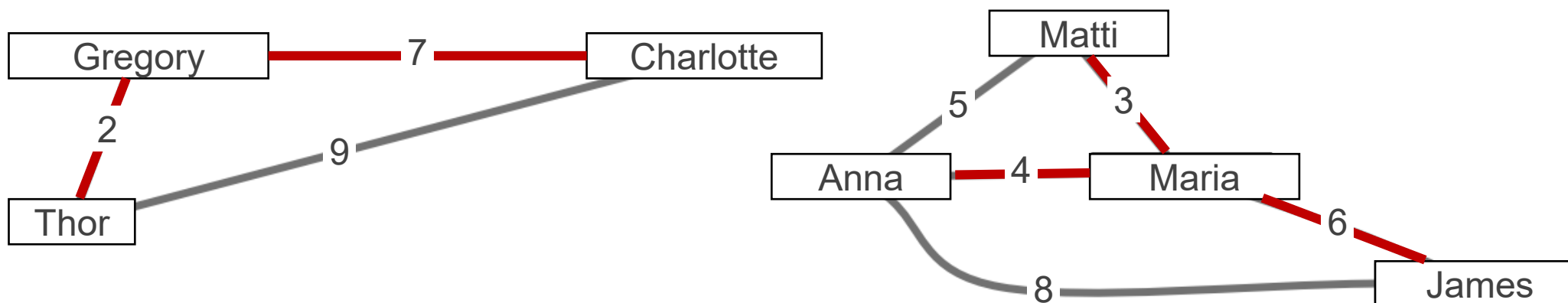
I want to take to the manager!



Neither algorithm is guaranteed to find the smallest solution.



MSF: Grow Algorithm



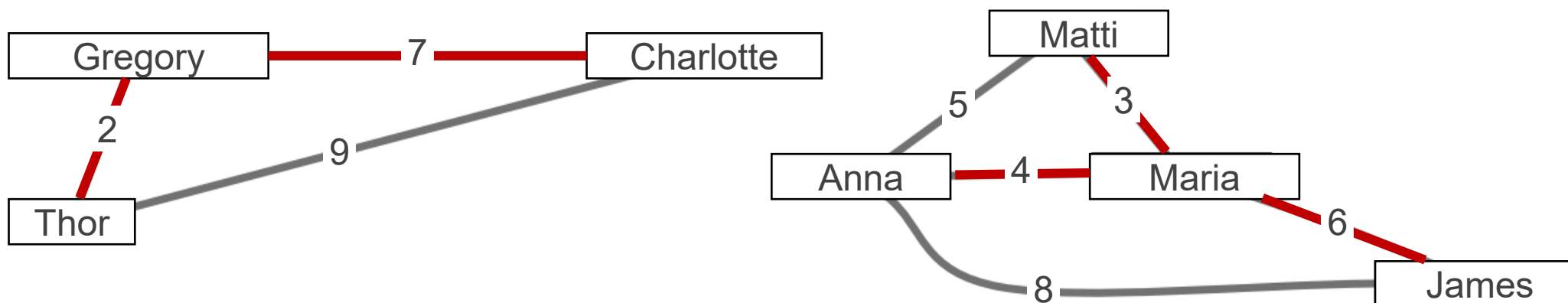
```
def GROW( $G$ )
   $S := \emptyset$ 
  consider the edges in increasing order
  for each edge  $e$ :
    if  $e$ 's endpoints are not yet connected
      add  $e$  to  $S$ .
```



Increasing order: 2, 3, 4, 5, 6, 7, 8, 9.



MSF: Shrink Algorithm



```
def SHRINK( $G$ )
```

$S = \{\text{all edges}\}$

consider the edges in order, from **highest-weight** to lowest-weight
for each edge e :

if every pair of nodes are connected via $S - \{e\}$:
remove e from S .



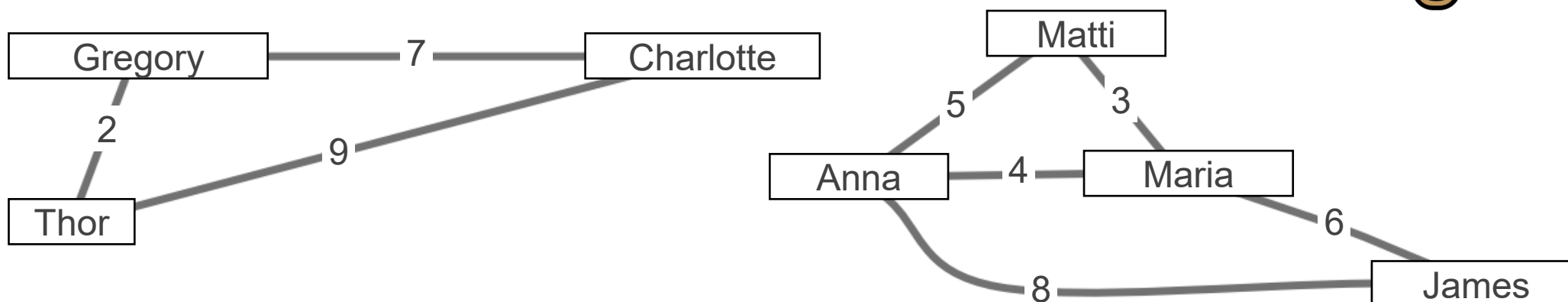
Decreasing order: 9, 8, 7, 6, 5, 4, 3, 2.



Contacts in the
study group



Minimum Spanning Forest (VI)



Minimum spanning forest problem:

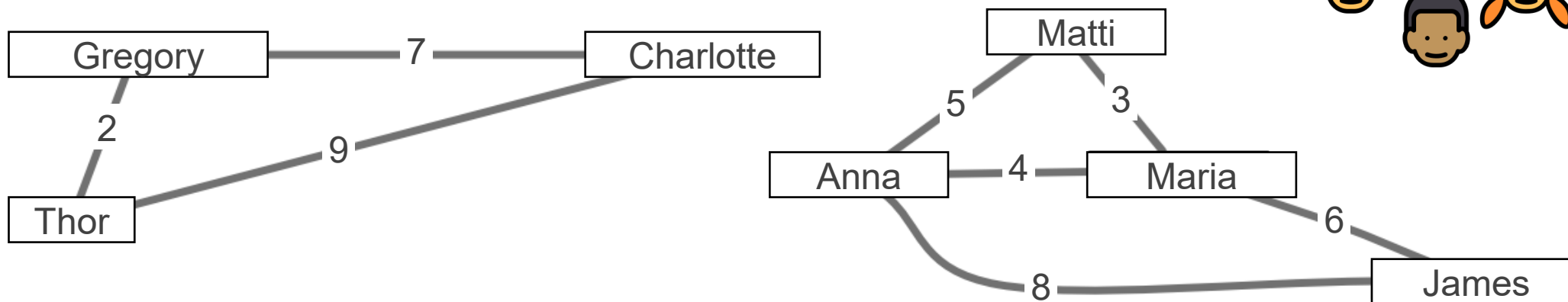
- ▶ *input:* a graph G , and an assignment of real-number weights to the edges of G .
- ▶ *output:* a minimum-weight set S of edges that is spanning and a forest.



Contacts in the
study group



Minimum Spanning Forest (VI)

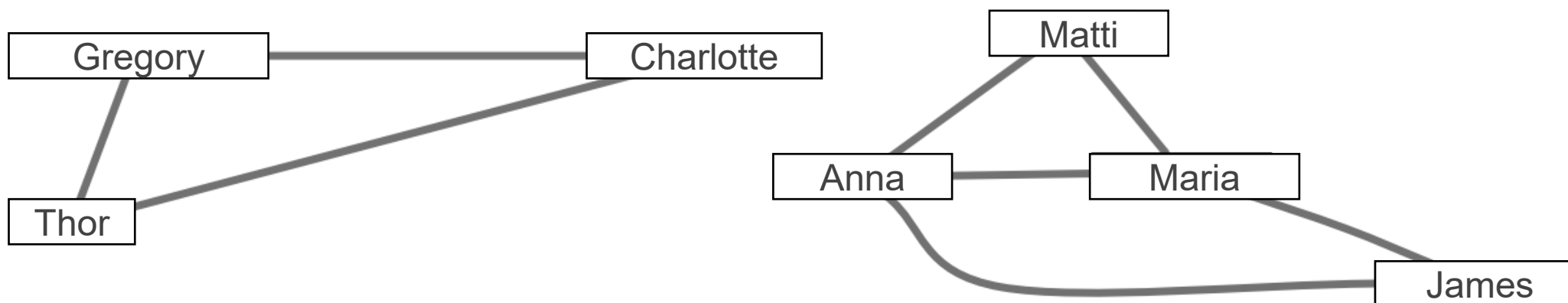


Application: Making sure that all students are informed about the study group's meeting

- ▶ Network must achieve same connectivity as input graph
- ▶ An edge represents a person getting in touch with another person (neighboring node)
- ▶ Weight of edge represents the **time needed (=cost) to convince the other to join the meeting**
- ▶ Goal: **minimize the time to arrange the meeting** (= minimize total cost)



MSF in Linear Algebra (I)



Let $D =$ set of nodes $\{\text{Gregory, Charlotte, Thor, Matti, Anna, Maria, James}\}$

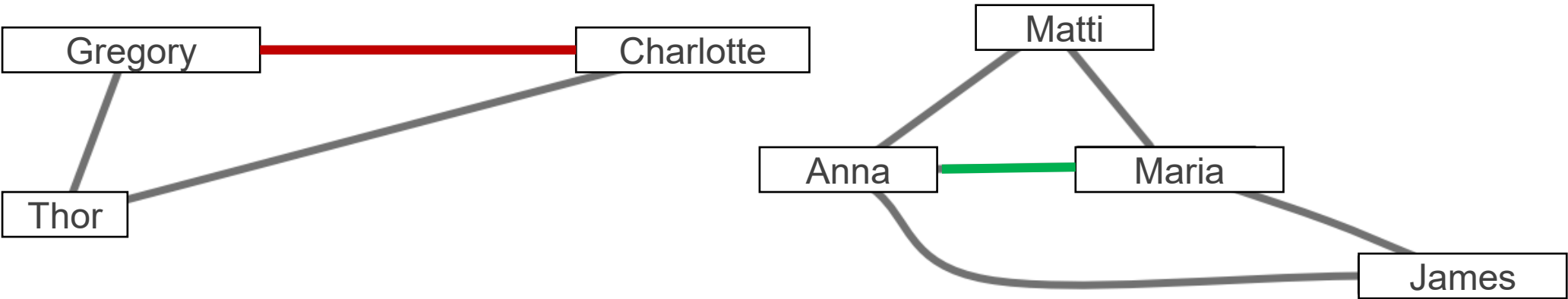
Represent a subset of D by a $GF(2)$ vector:

subset $\{\text{Gregory, Matti, James}\}$ is represented by

Gregory	Charlotte	Thor	Matti	Anna	Maria	James
1			1			1



MSF in Linear Algebra (II)

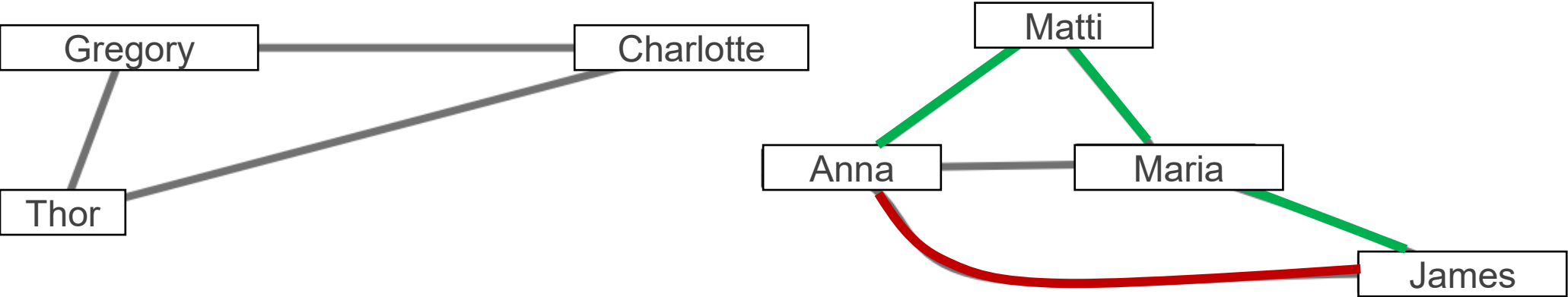


edge	vector						
	Gregory	Charlotte	Thor	Matti	Anna	Maria	James
{Gregory, Charlotte}	1	1					
{Gregory, Thor}	1		1				
{Charlotte, Thor}		1	1				
{Matti, Anna}				1	1		
{Matti, Maria}				1		1	
{Anna, Maria}					1	1	
{Anna, James}					1		1
{Maria, James}						1	1

Individual connections



MSF in Linear Algebra (III)



The vector representing $\{Anna, James\}$

Gregory	Charlotte	Thor	Matti	Anna	Maria	James
				1		1

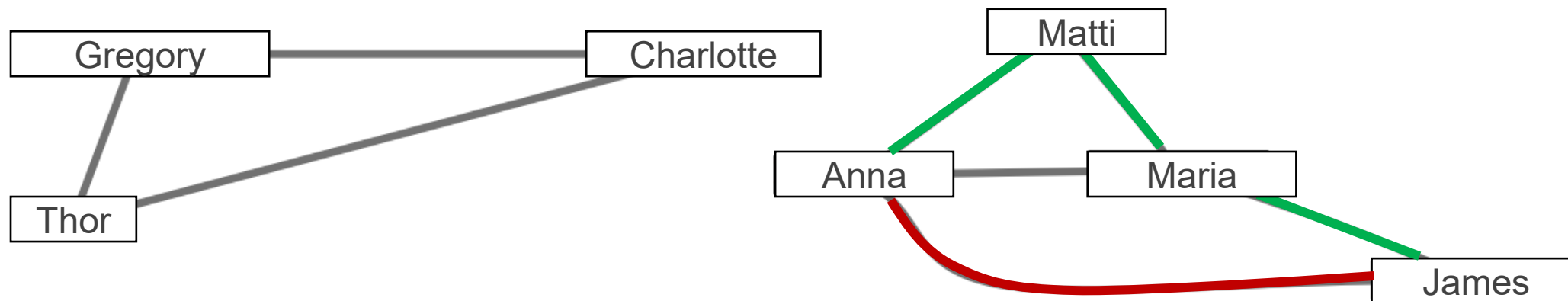
is the sum, for example, of the vectors representing $\{Anna, Matti\}$, $\{Matti, Maria\}$, and $\{Maria, James\}$

Gregory	Charlotte	Thor	Matti	Anna	Maria	James
			1	1		
			1		1	
					1	1
			0	1	0	1

GF(2)



MSF in Linear Algebra (IV)



A vector with 1's in entries x and y is the sum of vectors corresponding to edges that form an x -to- y path in the graph.

~~$\{Anna, Matti\}, \{Matti, Maria\}, \{Maria, James\}$~~

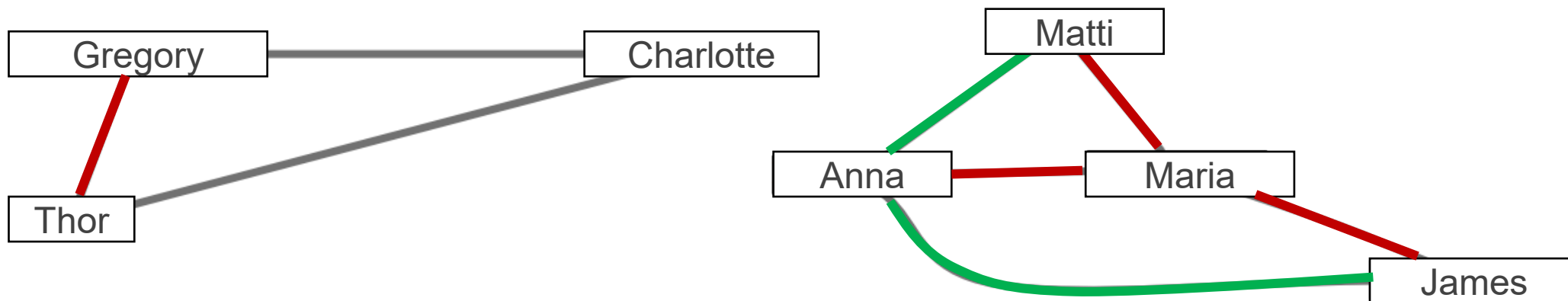
OR

~~$\{Anna, Maria\}, \{Maria, James\}$~~

Cancel each other out



MSF in Linear Algebra (V)



Example: The span of the vectors representing

$\{\text{Gregory, Thor}\}, \{\text{Matti, Maria}\}, \{\text{Anna, Maria}\}, \{\text{Maria, James}\}$

- ▶ contains the vectors corresponding to $\{\text{Matti, Anna}\}, \{\text{Anna, James}\},$ and $\{\text{Maria, James}\}$
- ▶ but not the vectors corresponding to e.g., $\{\text{Charlotte, Thor}\}, \{\text{Gregory, Charlotte}\},$ or $\{\text{Thor, Matti}\}$

Has again no one reached out to Charlotte about our meeting ?





MSF in Linear Algebra (VI)

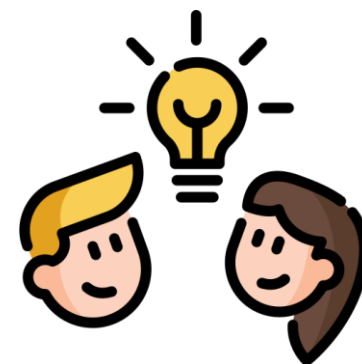
Using edges (graphs)

```
def GROW( $G$ )  
   $S := \emptyset$   
  consider the edges in increasing order  
  for each edge  $e$ :  
    if  $e$ 's endpoints are not yet connected  
      add  $e$  to  $S$ .
```

Using vectors

```
def GROW( $\mathcal{V}$ )  
   $S = \emptyset$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $\mathcal{V}$  not in  $\text{Span } S$ ,  
    and put it in  $S$ .
```

- ▶ Considering edges e of G corresponds to considering vectors $\mathbf{v}$ in $\mathcal{V}$
- ▶ Testing if e 's endpoints are not connected corresponds to testing if $\mathbf{v}$ is not in $\text{Span } S$.



Not in span = is not “contained” in / cannot be “reached” via the other vectors





Linear Dependence (III)



Different than the previous example

Example: The vectors $[1, 0, 0]$, $[0, 2, 0]$, and $[0, 0, 4]$ are linearly independent.

How do we know?

Easy since each vector has a nonzero entry where the others have zeroes.

Consider any linear combination

$$\alpha_1 [1, 0, 0] + \alpha_2 [0, 2, 0] + \alpha_3 [0, 0, 4]$$

This equals $[\alpha_1, 2\alpha_2, 4\alpha_3]$

If this is the zero vector, it must be that $\alpha_1 = \alpha_2 = \alpha_3 = 0$

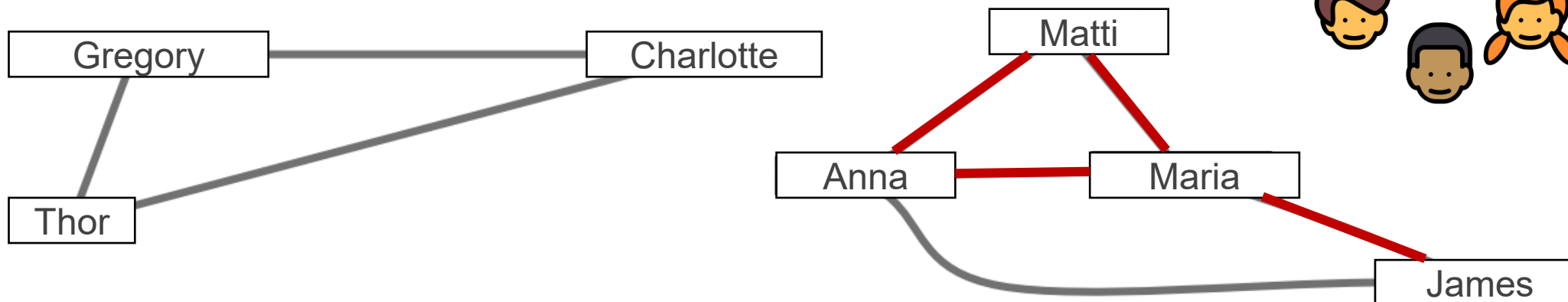
That is, the linear combination is trivial.

We have shown the only linear combination that equals the zero vector is the trivial linear combination.



Linear Dependence - MSF (I)

Contacts in the
study group



We can get the zero vector by adding together vectors corresponding to edges that form a cycle: in such a sum, for each entry x , there are exactly two vectors having 1's in position x .

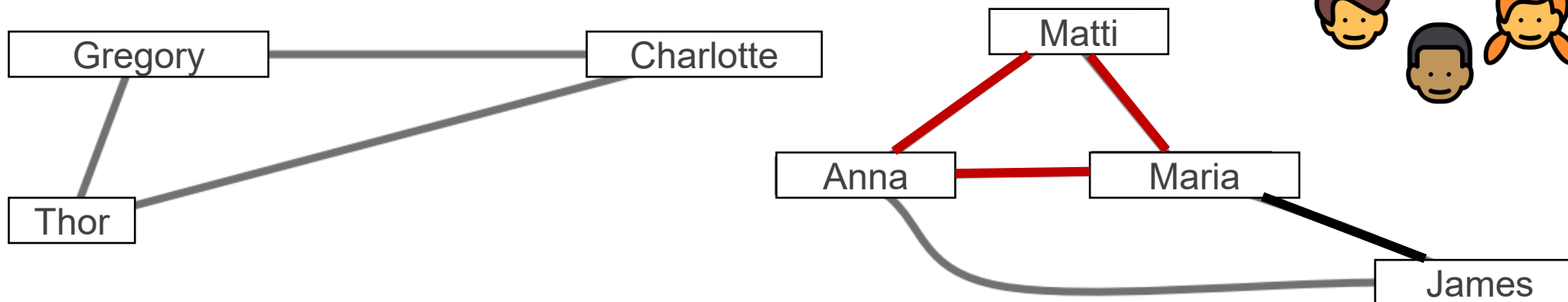
Therefore if a subset of S form a cycle then S is linearly dependent.

Having paths that result in a zero vector ($\rightarrow$ cycle) is like having nonzero α coefficients for vectors and still obtaining the zero vector



Linear Dependence - MSF (II)

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Example: the vectors corresponding to

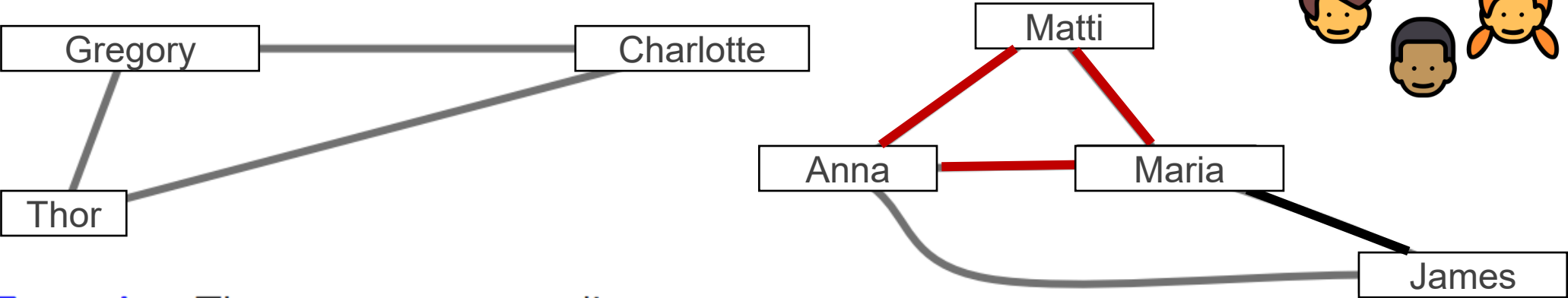
$\{\text{Matti, Maria}\}, \{\text{Matti, Anna}\}, \{\text{Anna, Maria}\}$

are as follows:

Gregory	Charlotte	Thor	Matti	Anna	Maria	James	
			1	1			
				1	1		
			1		1		
			0	0	0		
							GF(2)

Linear Dependence - MSF (III)

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Example: The vectors corresponding to $\{\text{Matti, Maria}\}, \{\text{Matti, Anna}\}, \{\text{Anna, Maria}\}, \{\text{Maria, James}\}$ are linearly dependent because these edges include a cycle.

The zero vector is equal to the nontrivial linear combination

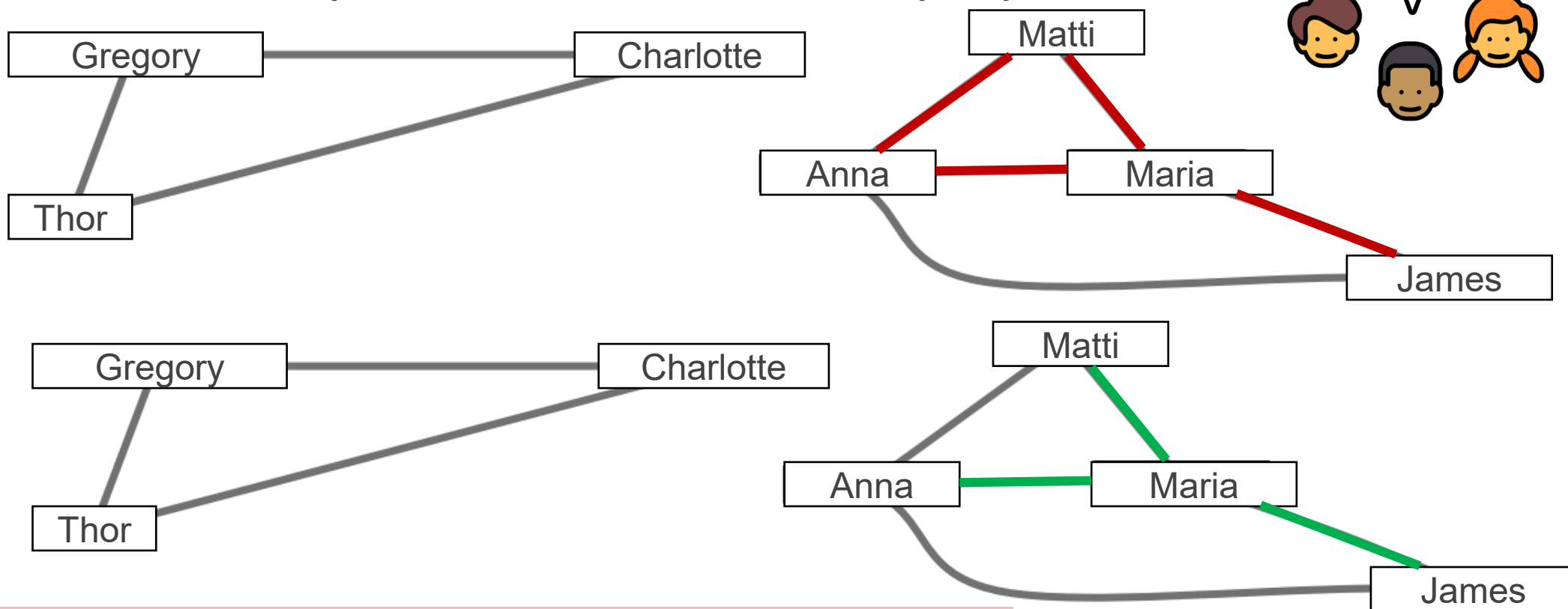
**Nonzero
coefficients
lead to zero
vector**

	↓		Gregory	Charlotte	Thor	Matti	Anna	Maria	James
	1	*				1	1		
+	1	*				1		1	
+	1	*					1	1	
+	0	*						1	1



Linear Dependence - MSF (IV)

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If a subset of S forms a cycle then S is linearly dependent.

On the other hand, if a set of edges contains no cycle (i.e. is a forest) then the corresponding set of vectors is linearly independent.



Loss Compression (I)

Say you need to store or transmit many 2-megapixel images:

How do we represent the image compactly?

- ▶ *Obvious method:* 2 million pixels $\implies$ 2 million numbers

- ▶ *Strategy 1:* Use sparsity! Find the “nearest” k -sparse vector. Later we’ll see this consists of suppressing all but the largest k entries.

- ▶ *More sophisticated strategy?*

Essentially setting many entries to zero (black)

Original



Strategy 1





Loss Compression (II)

Remember (few slides ago)

“ $\mathbf{u}$ is the coordinate representation of $\mathbf{v}$ in terms of $\mathbf{a}_1, \dots, \mathbf{a}_n$ ” can be written as matrix-vector equation $A\mathbf{u} = \mathbf{v}$

Strategy 2: Represent image vector by its coordinate representation:

- ▶ Before compressing any images, select vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$.
- ▶ Replace each image vector with its coordinate representation in terms of $\mathbf{v}_1, \dots, \mathbf{v}_n$.



Loss Compression (III)

Strategy 2: Represent image vector by its coordinate representation:

- ▶ Before compressing any images, select vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$.
- ▶ Replace each image vector with its coordinate representation in terms of $\mathbf{v}_1, \dots, \mathbf{v}_n$.

For this strategy to work, we need to ensure that every image vector can be represented as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$.

Given some D -vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ over $\mathbb{F}$, how can we tell whether *every* vector in $\mathbb{F}^D$ can be written as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$?

We also need the number of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ to be much smaller than the number of pixels.

Given D , what is minimum number of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ such that every vector in $\mathbb{F}^D$ can be written as a linear combination?



Loss Compression (IV)

Strategy 3: A hybrid approach

Step 1: Select vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$.

Step 2: For each image to compress, find its coordinate representation $\mathbf{u}$ in terms of $\mathbf{v}_1, \dots, \mathbf{v}_n$

Step 3: Replace $\mathbf{u}$ with the closest k -sparse vector $\tilde{\mathbf{u}}$, and store $\tilde{\mathbf{u}}$. **Sparsity-part**

Step 4: To recover an image from $\tilde{\mathbf{u}}$, calculate the corresponding linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$.



Loss Compression (V)

Strategy 1: Closest Sparse Representation



Strategy 3: Hybrid Approach





Analyzing the Grow Algorithm (I)

```
def GROW( $\mathcal{V}$ )  
   $S = \emptyset$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $\mathcal{V}$  that is not in  $\text{Span } S$ , and put it in  $S$ .
```

Grow-Algorithm Corollary: The vectors obtained by the Grow algorithm are linearly independent.

In graphs, this means that the solution obtained by the Grow algorithm has no cycles (is a forest).





Analyzing the Grow Algorithm (II)



Proof: For $n = 1, 2, \dots$, let $\mathbf{v}_n$ be the vector added to S in the n^{th} iteration of the Grow algorithm. We show by induction that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

For $n = 0$, there are no vectors, so the claim is trivially true.

Assume the claim is true for $n = k - 1$. We prove it for $n = k$.

The vector $\mathbf{v}_k$ added to S in the k^{th} iteration is not in the span of $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$.

Therefore, by the Linear-Dependence Lemma, for any coefficients $\alpha_1, \dots, \alpha_k$ such that

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_{k-1} \mathbf{v}_{k-1} + \alpha_k \mathbf{v}_k$$

it must be that α_k equals zero. We may therefore write

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_{k-1} \mathbf{v}_{k-1}$$

By claim for $n = k - 1$, $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ are linearly independent, so $\alpha_1 = \dots = \alpha_{k-1} = 0$

The linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$ is *trivial*. We have proved that $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent. This proves the claim for $n = k$. QED



Analyzing the Shrink Algorithm (I)

```
def SHRINK( $\mathcal{V}$ )  
   $S$  = some finite set of vectors that spans  $\mathcal{V}$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $S$  such that  $\text{Span}(S - \{\mathbf{v}\}) = \mathcal{V}$ , and remove  $\mathbf{v}$  from  $S$ .
```

↑
Remove superfluous vectors
↑

Remember (some slides ago)

Superfluous-Vector Lemma For any set S and any vector $\mathbf{v} \in S$, if $\mathbf{v}$ can be written as a linear combination of the other vectors in S then
 $\text{Span}(S - \{\mathbf{v}\}) = \text{Span } S$



Analyzing the Shrink Algorithm (II)

```
def SHRINK( $\mathcal{V}$ )  
   $S$  = some finite set of vectors that spans  $\mathcal{V}$   
  repeat while possible:  
    find a vector  $\mathbf{v}$  in  $S$  such that  $\text{Span}(S - \{\mathbf{v}\}) = \mathcal{V}$ , and remove  $\mathbf{v}$  from  $S$ .
```

Shrink-Algorithm Corollary: The vectors obtained by the Shrink algorithm are linearly independent.

In graphs, this means that the Shrink algorithm outputs a solution that is a forest.





Analyzing the Shrink Algorithm (III)



Proof: Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be the set of vectors obtained by the Shrink algorithm. Assume for a contradiction that the vectors are linearly dependent.

Then $\mathbf{0}$ can be written as a nontrivial linear combination

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$$

where at least one of the coefficients is nonzero.

Let α_i be one of the nonzero coefficients.

By the Linear-Dependence Lemma, $\mathbf{v}_i$ can be written as a linear combination of the other vectors.

Hence by the Superfluous-Vector Lemma, $\text{Span}(S - \{\mathbf{v}_i\}) = \text{Span } S$,

so the Shrink algorithm should have removed $\mathbf{v}_i$.

QED